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The First Stars: Formation, Properties, and Impact

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Keywords

cosmology, first and second stellar population, galactic archeology, high-redshift Universe, Population III and II, star formation

Abstract

The first generation of stars, often called Population III (or Pop III), form from metal-free primordial gas at redshifts $z \sim 30$ and below. They dominate the cosmic star-formation history until $z \sim 15$–20, at which point the formation of metal-enriched Population II stars takes over. We review current theoretical models for the formation, properties, and impact of Pop III stars and discuss existing and future observational constraints. Key takeaways from this review include the following:

- Primordial gas is highly susceptible to fragmentation and Pop III stars form as members of small clusters with a logarithmically flat mass function.
- Feedback from massive Pop III stars plays a central role in regulating subsequent star formation, but major uncertainties remain regarding its immediate impact.
- In extreme conditions, supermassive Pop III stars can form, reaching masses of several $10^5\,M_\odot$. Their remnants may be the seeds of the supermassive black holes observed in high-redshift quasars.
- Direct observations of Pop III stars in the early Universe remain extremely challenging. Indirect constraints from the global 21-cm signal or gravitational waves are more promising.
- Stellar archeological surveys allow us to constrain both the low-mass and the high-mass ends of the Pop III mass distribution. Observations suggest that most massive Pop III stars end their lives as core-collapse supernovae rather than as pair-instability supernovae.

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1. INTRODUCTION

Population III stars (Pop III): stars formed from truly metal-free primordial gas in the high-redshift Universe

Primordial gas: Big Bang nucleosynthesis produces mostly H and ^{4}He with mass fractions of 0.76 and 0.24, respectively

Population II stars (Pop II): stars built up from metal-enriched material; early Pop II stars constitute the second stellar generation in cosmic history

From studying the cosmic microwave background (CMB), we know our Universe started out very simple. It was by and large homogeneous and isotropic, with small fluctuations that can be described by linear perturbation theory. In stark contrast, the Universe today is highly structured on a vast range of length and mass scales. In the evolution toward increasing complexity, the formation of the first stars marks a primary transition phase of cosmic evolution. Their light ends the so-called dark ages, and they play a key role in cosmic metal enrichment and reionization, thereby shaping the Universe at large and the present-day galaxy population. The study of stellar birth in the early Universe is therefore important for many areas of modern astronomy and astrophysics but is also a relatively young field of research. Only with the advent of advanced numerical methods and powerful supercomputers did a comprehensive modeling of early star formation become feasible. As a consequence, there is still considerable debate about the physical processes that govern stellar birth at high redshifts and the overall properties of the first stars. This review aims at providing an overview of the current state of the field.

The first generation of stars, the so-called Population III stars (Pop III) build up from truly metal-free primordial gas. Second generation stars, sometimes termed early Population II stars (Pop II), form from material that has been enriched from the debris of the first stars. Note that there are also Population I stars, such as the Sun, which are relatively young and metal rich. They are mostly found in the thin disk of the Milky Way and other spiral galaxies. Unlike the very first stars, which have not yet been directly detected, members of the second generation have been

found and characterized in surveys looking for extremely metal-poor (EMP) stars in our Milky Way and neighboring satellite galaxies. In concert with observational data at high redshift, this allows us to constrain the properties of genuine Pop III stars. Initially, primordial star formation was proposed to be very simple, governed by well-defined initial conditions provided by Gaussian fluctuations of the cosmic density field. These were thought to result in the buildup of solitary high-mass stars. This simple picture has undergone dramatic revisions. We now understand that fragmentation is a wide-spread phenomenon during first star formation, and that Pop III stars form as members of multiple stellar systems with separations as small as the distance between the Earth and the Sun. Studies that include radiative feedback, magnetic fields, dark matter annihilation, as well as the primordial streaming velocities add to this complexity. All of these processes are relevant and need to be included in any realistic model. There is agreement now that primordial star formation is just as dynamic and complicated as stellar birth at the present day.

We structure our review as follows: First, we provide in Section 2 an overview of all relevant physical processes that contribute to first star formation. This includes an introduction of the cosmological model that constitutes the basis of our analysis, together with the equations that govern the evolution of the cosmic fluid. We end this part with an account of the instabilities that lead to gravitational collapse and subsequently to stellar birth, and we stress the importance of the thermodynamic properties of the star-forming gas. In Section 3, we discuss the critical mass for collapse in the primordial Universe and describe the evolution of the cosmic star-formation rate density (SFRD). We then discuss in detail the standard Pop III formation pathway in isolated pristine halos. We also account for more complex scenarios in which further physical processes add to the complexity of the problem. Finally, we turn our attention to the most extreme physical conditions that result in the formation of supermassive stars (SMSs) and lead to the supermassive black holes (SMBHs) we observe in quasars at high redshift. The impact of stellar feedback is the focus of Section 4. We first look at the consequences of the radiative and the mechanical energy and momentum input from young massive stars on their immediate birth environment and then consider their influence on neighboring halos and on the larger-scale intergalactic gas. We also discuss chemical feedback of Pop III stars, which drives the transition to Pop II star formation and governs the early metal enrichment of the Universe. A critical assessment of possible observational probes of primordial star formation is presented in Section 5, covering measurements at high redshift as well as multimessenger data from the local Universe. Finally, we conclude and summarize in Section 6.

Our focus here is to provide a comprehensive overview of the developments and successes in the field during the past one or two decades. For further reading, including a more historic account, we refer to the reviews by Barkana & Loeb (2001), Bromm & Larson (2004), Glover (2005), Yoshida et al. (2012), and Bromm (2013), as well as to the book by Loeb (2010). A good overview with a specific focus on astrochemistry is provided by Glover (2005, 2013), and an account of important numerical aspects of high-redshift star formation is given by Greif (2015). For further reading on the transition to the second generations of stars and buildup of the first galaxies, we recommend Bromm & Yoshida (2011) or the textbook by Loeb & Furlanetto (2013). In order to calibrate our understanding of primordial star formation with what we know about stellar birth at present days, we also refer to the reviews by Mac Low & Klessen (2004), McKee & Ostriker (2007), Zinnecker & Yorke (2007), Krumholz (2015), or Klessen & Glover (2016) on different aspects of this subject.

2. IMPORTANT PHYSICAL CONCEPTS

In this section, we introduce the basic physical concepts needed to understand the formation of the first stars. We begin with the cosmological model that forms the foundation of all further

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considerations, introduce the main equations governing the evolution of the cosmic fluid, and study the criteria for the onset of gravitational instability in the early Universe and consequently for the formation of the first stars.

2.1. Cosmological Model

Throughout this review, our approach is based on the standard ΛCDM model (e.g., Peebles 1993, Bullock & Boylan-Kolchin 2017), in which the Universe consists of matter in the form of cold dark matter and baryons, and radiation, as well as dark energy. In this model, the Hubble parameter $H(a) = \dot{a}/a$, which describes the expansion of the Universe, evolves as

$$\frac{H^2(a)}{H_0^2} = \frac{\Omega_r}{a^4} + \frac{\Omega_\mathrm{m}}{a^3} + \frac{\Omega_\lambda}{a^{3(1+w)}} \quad \text{with} \quad 1 = \Omega_\mathrm{r} + \Omega_\mathrm{m} + \Omega_\lambda, \qquad 1.$$

where $a(z) = (1+z)^{-1}$ is the cosmic scale factor, and z is the redshift. At the present day, i.e., for $a = 1$, the densities of radiation, matter, and dark energy with respect to the critical density are $\Omega_\mathrm{r} \approx 10^{-5}$, $\Omega_\mathrm{m} = 0.31$, and $\Omega_\lambda = 0.69$, respectively (Planck Collab. et al. 2020b). The parameter w relates the pressure and energy of the dark energy component ($p = w\rho c^2$); here, we assume that it is constant in time and equal to -1, as appropriate for the classical cosmological constant. The current value[1] of the Hubble parameter H_0 is $\sim$70 km s^{-1}Mpc^{-1}; for historical reasons, this is usually expressed as $H_0 = 100\,h$ km s^{-1}Mpc^{-1} with $h \approx 0.7$.

2.2. Evolution of the Cosmic Fluid

In order to understand primordial star formation, we need to identify and characterize the first regions in the Universe that decouple from the cosmic expansion and contract under their own gravitational attraction. Observations of the CMB show that the Universe started out extremely simple. The initial density distribution was isotropic and almost perfectly homogeneous with spatial fluctuations, $\delta(\mathbf{x}) = \rho(\mathbf{x})/\langle\rho\rangle - 1$, of order 10^{-5} on large scales with respect to the average background density $\langle\rho\rangle$. In the dark matter, these fluctuations are present on all scales and grow due to their own self-gravity. In the linear regime, the evolution of δ is governed by the equation

$$\frac{\partial^2\delta}{\partial t^2} + 2H\frac{\partial\delta}{\partial t} = 4\pi G\langle\rho\rangle\delta. \qquad 2.$$

This corresponds to the famous Jeans (1902) equation describing the evolution of self-gravitating isothermal gas spheres, written in the limit of zero pressure, and with an additional drag term due to the expansion of the Universe. In the ΛCDM model, the initial density fluctuations are well characterized as a Gaussian random field, and so their statistical properties are completely determined by the power spectrum of the field, which has been inferred with high precision from the observed temperature fluctuations in the CMB (Planck Collab. et al. 2020a). Consequently, the physical properties of the Universe at a redshift of $z \sim 1{,}000$, and hence the initial conditions for cosmic structure formation, are extremely well constrained.

2.3. Gravitational Instability

Whereas (cold) dark matter can be considered a pressure-less zero-temperature fluid, this is not the case for the baryonic component of the Universe. On large scales, the gravity of the dark matter

[1]There remains some debate regarding the precise value of H_0—for a comprehensive overview, see Valentino et al. (2021).

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dominates, and the evolution of the baryons is very similar to that of the dark matter. On small scales, the effects of gas pressure become increasingly important. In the classical Jeans (1902) stability analysis, we can identify a critical mass scale M_J separating the gravity-dominated and pressure-dominated regimes. Perturbations with $M > M_J$ undergo gravitational collapse, whereas those with $M < M_J$ do not. In the simple case of an isothermal gas sphere, we have

$$M_J = \frac{\pi^{5/2}}{6}\left(\frac{1}{G}\right)^{3/2}\rho^{-1/2}c_s^3 \approx 50\,M_\odot\,\mu^{-2}\left(\frac{n}{1\,\mathrm{cm}^{-3}}\right)^{-1/2}\left(\frac{T}{1\,\mathrm{K}}\right)^{3/2}, \quad 3.$$

where n is the number density of H nuclei and c_s is the sound speed, which are related to T and ρ by $\rho = \mu m_H n$ and $T = \mu m_H c_s^2/k_B$. Information about the chemical makeup of the gas is encoded in the weight factor $\mu = (1 + 4\chi)$ with χ being the ratio of He to H atoms by number. For primordial gas, $\chi = 0.079$ and so $\mu = 1.32$. Finally, we note that a similar expression to Equation 3 can be derived by considering the growth of plane-wave baryonic perturbations in the linear regime (see, e.g., Peacock 1999). More generally, an expression differing only by a small numerical factor can be derived by comparing the gravitational and sound-crossing timescales or the gravitational and thermal energy of the perturbation.

Equation of state: relates thermodynamic state variables such as pressure, density, temperature, or internal energy, and is needed to turn the equations of hydrodynamics into a closed (and thus solvable) system

For a collapsing sphere, we can estimate the associated accretion rate from the Jeans mass and the free-fall time $\tau_{ff} = (3\pi/32G\rho)^{1/2}$ via

$$\dot{M} = f\frac{M_J}{\tau_{ff}} \approx f\frac{c_s^3}{G}, \quad 4.$$

where f can take on values between ~1 and several tens, depending on the initial density profile and the ratio between M and M_J. Note, in this simple approximation, the accretion rate only depends on the gas temperature. For further discussion, see Whitworth & Summers (1985) and references therein. Additional physical processes can also be accounted for in our definition of the Jeans mass, most easily by defining and using an effective sound speed in place of c_s. For example, when the gas is turbulent on scales much smaller than the dynamical scales of interest (Chandrasekhar 1951a,b; von Weizsäcker 1951), we can simply add the velocity dispersion σ to the sound speed. A similar role has been ascribed to the magnetic field B, in which case we include a contribution from the Alfvén velocity, $v_A = B/(8\pi\rho)^{1/2}$ (e.g., Federrath & Klessen 2012), leading to

$$c_{s,\mathrm{eff}} = \left(c_s^2 + \sigma^2 + \frac{1}{2}v_A^2\right)^{1/2}. \quad 5.$$

We revisit these aspects later in Section 3.3.

2.4. Impact of Thermodynamics

The above considerations demonstrate the importance of the thermodynamic response of the evolving system (see also Section 3.2.1). It is best described by the equation of state, which relates thermodynamic state variables, such as pressure, density, temperature, or internal energy, and which is needed to turn the equations of hydrodynamics into a closed and thus solvable system. As an illustration, look at the simplified case of the gas following an effective polytropic equation of state,

$$p \propto \rho^{\gamma_{\mathrm{eff}}} \quad \text{with index} \quad \gamma_{\mathrm{eff}} = 1 + \mathrm{d}\ln T/\mathrm{d}\ln\rho, \quad 6.$$

where the γ_{eff} is the result of the competition between various heating and cooling mechanisms (e.g., Omukai et al. 2005, Klessen & Glover 2016). From Equation 3, we see that

$$M_J \propto \rho^{\frac{3}{2}(\gamma_{\mathrm{eff}}-\frac{4}{3})}. \quad 7.$$

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If $\gamma_{\rm eff} > 4/3$, the Jeans mass increases during the contraction and eventually becomes comparable with the mass of the system (including both dark matter and baryons). In this case, pressure forces stop further collapse. For adiabatic gas with $\gamma_{\rm eff} = 5/3$, this happens long before stellar densities are reached. In order for stars to form, the gas must be able to radiate energy away during its collapse, so that $\gamma_{\rm eff}$ remains below the critical value.

The importance of cooling can be assessed by comparing the cooling time

$$\tau_{\rm cool} = \frac{1}{\gamma - 1} \frac{n_{\rm tot} k_{\rm B} T}{\Lambda(T, n)}, \qquad 8.$$

with the free-fall timescale, $\tau_{\rm ff}$. Here, $n_{\rm tot}$ is the total number density of particles, $k_{\rm B}$ is the Boltzmann constant, T is the temperature, and Λ is the cooling rate. If $\tau_{\rm cool} < \tau_{\rm ff}$ then the gas can cool rapidly, and collapse proceeds roughly on a free-fall timescale. However, if $\tau_{\rm ff} < \tau_{\rm cool}$, then the gas quickly becomes pressure supported, and the contraction slows down and proceeds quasistatically for a duration on the order of $\tau_{\rm cool}$.

2.5. Instability of Rotationally Supported Systems

In rotationally supported systems, the criterion for gravitational instability takes a slightly different form, owing to the stabilizing effect of shear. For infinitely thin disks, this was investigated by Toomre (1964), who derived the following criterion for instability:

$$Q = \frac{\kappa c_{\rm s}}{\pi G \Sigma} \lesssim 1. \qquad 9.$$

Here, Σ and κ are the surface density and epicyclic frequency, respectively. For systems in Keplerian rotation, we have $\kappa = \Omega$, where Ω is the rotational frequency (Kratter & Lodato 2016). This approach can be extended to thick disks with multiple components (Rafikov 2001, Elmegreen 2002, Romeo & Falstad 2013) by introducing appropriate correction factors to Equation 9. There are two main pathways toward disk fragmentation. First, in the absence of accretion onto the system, Gammie (2001) argues that even an initially stable disk becomes unstable if the cooling timescale, $\tau_{\rm cool}$, is shorter than the orbital timescale, $\tau_{\rm orbit} = 1/\Omega$. Second, in the presence of accretion from the surrounding gas envelope, if the mass load onto the disk exceeds its capability to transport material inward, Σ increases beyond the critical value and the disk becomes unstable. As we see later, this latter scenario is commonly encountered when studying Pop III accretion disks.

3. POPULATION III STAR FORMATION

We begin our discussion of Pop III star formation with a critical review of their birth environment, then turn our attention to the most likely formation pathway and speculate about alternative scenarios. In the most extreme cases this can lead to the formation of SMSs as possible progenitors of SMBHs.

3.1. Critical Mass for Collapse and Cosmic Star-Formation Rate Density

The formation of the first stars in the Universe occurs in regions where the cosmic fluid fulfils two conditions. First, it needs to decouple from the global expansion and begin to contract due to the self-gravity of dark matter. We call a region where this happens a dark matter halo. Second, the gas within the dark matter halo needs to be able to cool and go into runaway collapse to dramatically increase the baryon–to–dark matter ratio and eventually reach stellar densities. As we discuss in more detail later, halos with virial temperatures above $T_{\rm vir} \sim 8{,}000$ K can cool initially via Lyman-α emission from atomic hydrogen, whereas lower-mass halos with lower $T_{\rm vir}$ depend

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on cooling from H_2. For historical reasons, the latter type of halo is often referred to as a minihalo. In the current ΛCDM paradigm, gravitationally bound objects form in a hierarchical fashion with smaller objects forming first, implying that minihalos are the first sites in which Pop III stars can potentially form.

Streaming velocity: second-order cosmic perturbation theory predicts a relative motion between dark matter and baryons, which decreases linearly as the Universe expands

3.1.1. Simple models for the critical mass. Calculations of the growth of density perturbations in an expanding Universe (e.g., Barkana & Loeb 2001) show that substantial baryonic overdensities can develop only in halos with masses above a critical mass given by

$$M_{\rm crit} \approx 5 \times 10^3 {\rm M}_\odot \left(\frac{\Omega_{\rm m} h^2}{0.143}\right)^{-1/2} \left(\frac{\Omega_{\rm b} h^2}{0.022}\right)^{-3/5} \left(\frac{1+z}{10}\right)^{3/2}. \quad 10.$$

This mass scale is very similar to the Jeans mass (Equation 3) of fluctuations, with a density close to the mean density of the Universe and a temperature that declines adiabatically as the Universe expands, and holds for $z \lesssim 100$ when gas is no longer thermally coupled to the CMB by Compton scattering. However, the development of a baryonic overdensity is a necessary but not sufficient condition for the formation of Pop III stars. In addition, as we have argued above, the gas must also be able to cool. This requirement yields a much higher critical mass. For instance, using a very simple model for the behavior of gas in a high-redshift minihalo, Glover (2013) derives a value

$$M_{\rm crit} \approx 1.4 \times 10^6 {\rm M}_\odot \left(\frac{\Omega_{\rm m} h^2}{0.143}\right)^{-1/2} \left(\frac{\mu}{1.32}\right)^{-3/2} \left(\frac{1+z}{10}\right)^{-3/2}, \quad 11.$$

where we encounter again the chemical weight factor μ introduced in Equation 3.

3.1.2. Models with more physics included. Clearly, Equations 10 and 11 are substantial simplifications. For example, the latter assumes that the chemical evolution of the gas takes place at constant density and temperature, and neither equation accounts for the fact that conditions in the Universe change during the time that it takes for a perturbation to grow into the nonlinear regime. This can lead to significant deviations from the prescriptions above (see, e.g., Gnedin & Hui 1998 or Naoz & Barkana 2007 for more complex models). To capture the nonlinear evolution and to account for the complex interplay between the different physical processes involved requires us to resort to numerical simulations. Processes that must be accounted for include the buildup of a cosmic Lyman–Werner (LW) or X-ray radiation background as star formation sets in (e.g., Gnedin 2000; see also Sections 4.3.2 and 4.3.3) and the existence of relative streaming velocities between dark matter and baryons (Tseliakhovich & Hirata 2010, Tseliakhovich et al. 2011; see also Section 3.3.1).

The latter results from second-order cosmological perturbation theory and therefore is often neglected in a purely linear analysis. Prior to recombination, baryons are tightly coupled to photons, resulting in a standing acoustic wave pattern (Sunyaev & Zel'dovich 1970) and consequently in oscillations between baryons and dark matter with relative velocities of about $30\,{\rm km\,s^{-1}}$ and coherence lengths of a few megaparsecs in comoving units at $z \approx 1{,}000$ (Silk 1968). After recombination, baryons are no longer tied to photons, their sound speed drops to $\sim 6\,{\rm km\,s^{-1}}$, and the velocity with respect to the dark matter component becomes supersonic with Mach numbers of $\mathcal{M} \approx 5$ (Tseliakhovich & Hirata 2010). As the Universe expands, the relative streaming velocity decays linearly and reaches $\sim 1\,{\rm km\,s^{-1}}$ at $z \approx 30$, which is comparable with the virial velocity of the first halos to cool and collapse (Fialkov et al. 2012). Simulations that include streaming velocities suggest that their presence reduces the baryon overdensity in low-mass halos, delays the onset of cooling, and leads to a larger critical mass for collapse to set in (Greif et al. 2011b; Maio et al. 2011; Stacy et al. 2011; Naoz et al. 2012, 2013; O'Leary & McQuinn 2012; Latif et al. 2014a; Schauer et al. 2017b; Nakazato et al. 2022).

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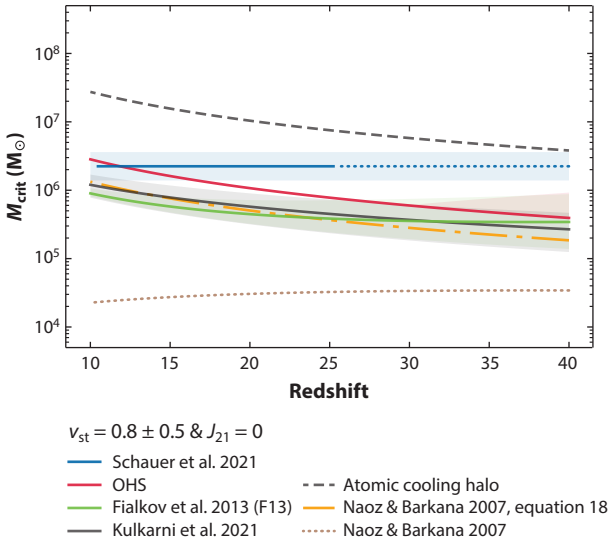

Figure 1

Critical mass for fluctuations in the cosmic fluid for collapse to set in. The solid lines show results from models that take the streaming velocity into account, adopted from Schauer et al. (2021), Kulkarni et al. (2021), and also models OHS and F13 from Hartwig et al. (2022) and Chen et al. (2022), which are derived from O'Shea & Norman (2008), Hummel et al. (2012), and Stacy et al. (2011) and from Fialkov et al. (2012), 2013), respectively. The fit from Schauer et al. (2021) is based on simulation data at $z < 25$, and we indicate our extrapolation of it to higher redshifts with a dotted line. The lines are computed for the most probable value of the streaming velocity, $v_{st} = 0.8$ (in units of the root-mean-squared value; see Schauer et al. 2019), and the shading indicates the effect of varying this value by ± 0.5. For further comparison, we show an estimate of M_{crit} in the absence of streaming based on our Equation 11 (*dash-dotted line*). The lower envelope (*dotted line*) shows the minimum halo mass required for a baryonic overdensity to develop, as computed by Naoz & Barkana (2007; see their equations 17 and 18). The upper envelope (*dashed line*) is given by the critical mass for atomic cooling halos (see, e.g., Hummel et al. 2012, their equation 1, or Hartwig et al. 2022, their equation 15). Abbreviations: OHS, O'Shea–Hummel–Stacy; F13, Fialkov et al. (2013).

They may also have substantial impact on the resulting 21-cm emission (Fialkov et al. 2012, McQuinn & O'Leary 2012, Visbal et al. 2012). Increasing the LW background intensity also increases M_{crit}, as discussed in more detail in Section 4.3.2.

Altogether, there remains significant uncertainty about the most appropriate value of M_{crit} with different simulations reaching different conclusions, depending on the numerical resolution adopted and the number of physical processes included (see, e.g., Yoshida et al. 2003, Latif & Schleicher 2019, Kulkarni et al. 2021, in addition to the above). We illustrate some of these variations in **Figure 1**, in which we plot predictions for M_{crit} from Schauer et al. (2021) and Kulkarni et al. (2021) and from models OHS and F13 in Hartwig et al. (2022) and Chen et al. (2022), which themselves are derived from the prescriptions of O'Shea & Norman (2008), Hummel et al. (2012), and Stacy et al. (2011) and from Fialkov et al. (2012, 2013), respectively. For further details, see appendix A.2 of Hartwig et al. (2022). The curves shown are computed in the absence of LW background radiation ($J_{21} = 0$), and for a streaming velocity $v_{st} = 0.8$ (in units of the root-mean-squared value), which is the most likely value to be encountered in the Universe (Schauer et al. 2019, 2021). For a discussion of the large-scale impact of LW feedback and the associated uncertainties, we again refer to Section 4.3.2. For completeness, we also plot the estimate for M_{crit}

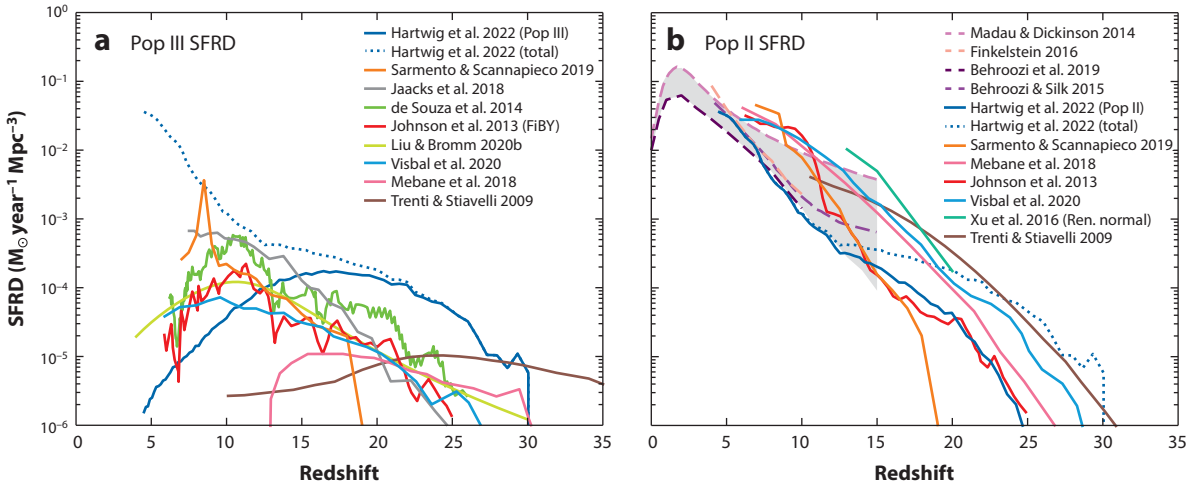

Figure 2

SFRD for Pop III and Pop II stars at $z > 5$ from different studies, including numerical simulations (Johnson et al. 2013, de Souza et al. 2014, Xu et al. 2016, Jaacks et al. 2018, Sarmento et al. 2019, Liu & Bromm 2020b) and semianalytical models (Trenti & Stiavelli 2009, Mebane et al. 2018, Visbal et al. 2020, Hartwig et al. 2022, Magg et al. 2022a). For comparison, we also provide the observational constraints from Madau & Dickinson (2014), Behroozi & Silk (2015), Finkelstein (2016), and Behroozi et al. (2019) with the gray shaded area indicating the proposed range. Abbreviations: FiBY, First Billion Years project; Pop II, Population II; Pop III, Population III; Ren., Renaissance simulation suite; SFRD, star-formation rate density.

with $v_{st} = 0$ given by Equation 11, the minimum mass required for the development of a baryonic overdensity computed by Naoz & Barkana (2007), and the critical mass for an atomic cooling halo (taken from Hummel et al. 2012 and Hartwig et al. 2022). These systems have a virial temperature of about 10^4 K, and their thermodynamic properties are dominated by Lyman-α cooling rather than H_2 cooling (Sutherland & Dopita 1993). Halos of this mass and above are able to cool and collapse even in the presence of a very strong LW radiation background (Oh & Haiman 2002) or a high streaming velocity (Schauer et al. 2019), and so this curve constitutes the upper envelope of all models for M_{crit} considered here.

3.1.3. Cosmic star-formation rate density. In the ΛCDM model, Pop III star formation begins at a redshift of $z \gtrsim 30$ in rare high-sigma fluctuations. The rate increases as collapse becomes possible in more and more halos and reaches a peak at redshifts $z \sim 15$–20. Although the overall cosmic star-formation rate continues to increase (Madau et al. 2014), the rate at which metal-free Pop III stars form declines again, because regions in the Universe that have not been enriched by supernova (SN) ejecta from massive stars become increasingly rare. This is the transition to the birth of (slightly) metal-enriched Pop II stars, which we discuss in Section 4.4. Different models for the SFRD as a function of redshift are depicted in **Figure 2**. The scatter gives a feeling for the current uncertainties in determining the onset and evolution of high-redshift star formation in both numerical simulations (Johnson et al. 2013, de Souza et al. 2014, Xu et al. 2016, Jaacks et al. 2018, Sarmento et al. 2019) and semianalytical models (Trenti & Stiavelli 2009, Mebane et al. 2018, Visbal et al. 2020, Hartwig et al. 2022). For comparison, we also provide the observational constraints from Madau & Dickinson (2014), Behroozi & Silk (2015), Finkelstein (2016), and Behroozi et al. (2019) with the total range indicated in gray, although we caution that the *James Webb Space Telescope* (JWST) will soon improve the situation for $z \gg 5$.

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3.2. Standard Population III Formation Pathway

We refer to stellar birth in pristine halos containing zero-metallicity gas, which has not been affected by stellar feedback from neighboring halos, as the standard Population III formation pathway. For early studies in this field see Yoneyama (1972), Hutchins (1976), Silk (1977a,b,c), Carlberg (1981), Kashlinsky & Rees (1983), Palla et al. (1983), Carr et al. (1984), or Stahler et al. (1986a,b). Here, we focus on the current state of affairs and briefly mention alternative scenarios in Section 3.3.

3.2.1. Initial collapse phase. As discussed above, the ability of the gas in a halo to collapse and form stars depends on its ability to cool. All main cooling processes in zero metallicity gas are related to hydrogen, either in atomic or molecular form. At high temperatures, collisions can populate the excited electronic states of H, which then de-excite by emitting Lyman series photons. This process is often referred to simply as Lyman-α cooling and is most efficient around temperatures of $\sim 10^4$ K. To reach lower temperatures, molecular hydrogen is needed. The lightness of the H_2 molecule and its lack of a dipole moment conspire to render it an ineffective coolant at very low temperatures: Its lowest-energy radiative transition is the $J = 2 \rightarrow 0$ transition in its vibrational ground state, with an energy that corresponds to a temperature of $\sim$512 K. The high-velocity tail in the thermal Maxwell–Boltzmann velocity distribution allows the gas to cool below this value, but only down to about 200 K (see Greif 2015). This is often termed the Pop III.1 formation pathway (e.g., McKee & Tan 2008, Clark et al. 2011a). The temperature can drop even further, if cooling by deuterated hydrogen (HD) takes over (Nagakura & Omukai 2005). HD has a nonzero dipole moment and its lowest-energy transition is between the $J = 1$ and $J = 0$ rotational levels, corresponding to a temperature of $\sim$128 K. This is sometimes called the Pop III.2 formation pathway, and in practice it becomes relevant only in regions with enhanced fractional ionization, for example, in very massive or in externally irradiated halos (see Sections 3.3 and 4.2).

For a full account of primordial chemistry and the associated cooling and heating processes, we refer to the reviews by Glover (2005, 2013), or Bovino & Galli (2019). Here, we focus on the most essential concepts. At low densities, H_2 has two main formation pathways. First, it can form by a two-stage reaction pathway involving the H^- ion as an intermediate step (McDowell 1961, Peebles & Dicke 1968). The reactions are

$$\mathrm{H} + \mathrm{e}^- \rightarrow \mathrm{H}^- + \gamma, \tag{12.}$$

$$\mathrm{H}^- + \mathrm{H} \rightarrow \mathrm{H}_2 + \mathrm{e}^-. \tag{13.}$$

Second, there is also a contribution from a similar reaction pathway involving H_2^+ as an intermediary molecule (Saslaw & Zipoy 1967):

$$\mathrm{H} + \mathrm{H}^+ \rightarrow \mathrm{H}_2^+ + \gamma, \tag{14.}$$

$$\mathrm{H}_2^+ + \mathrm{H} \rightarrow \mathrm{H}_2 + \mathrm{H}^+. \tag{15.}$$

Both reaction pathways require the gas to be partially ionized, and the amount of H_2 that can form via these routes is limited by the slow formation rates of the H^- and H_2^+ ions and the recombination of the gas. Typically, the final molecular fraction is around 10^{-3}. At high particle densities above $\sim 10^9\,\mathrm{cm}^{-3}$, a third process becomes important. It is the three-body reaction (Palla et al. 1983):

$$\mathrm{H} + \mathrm{H} + \mathrm{H} \rightarrow \mathrm{H}_2 + \mathrm{H}. \tag{16.}$$

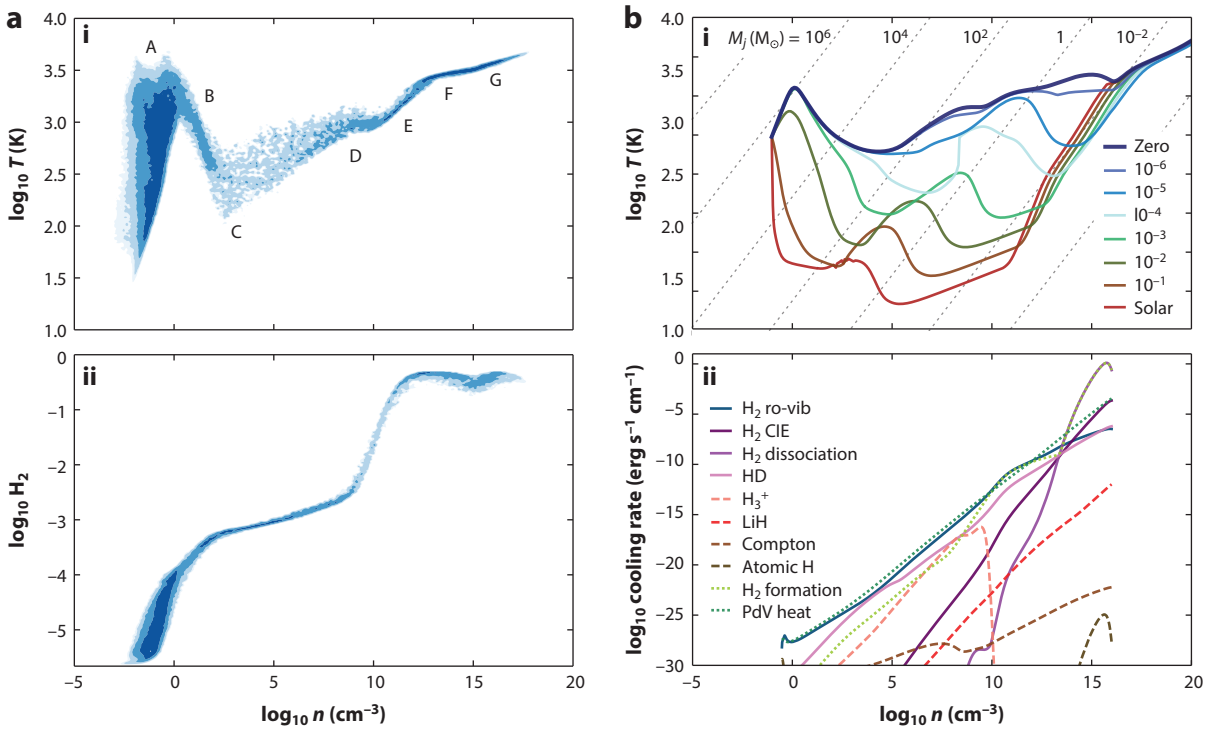

Figure 3

The relation between temperature T and hydrogen number density n during (*a*, *i*) the initial collapse phase and (*a*, *ii*) the corresponding fraction of molecular hydrogen. The data are taken from a resimulation of a halo from Schauer et al. (2021) provided by L. Prole. (*b*, *i*) The relation between $\log_{10}T$ and $\log_{10}n$, corresponding to the effective equation of state of the gas, for different metallicities, ranging from a purely primordial composition (*dark blue*, $Z = 0$) to the solar value (*red*, $Z = Z_\odot$) in steps of ten. The data are computed with a one-zone astrochemical model following the collapse of an isolated system (for full details, see Omukai et al. 2005, 2010). The dashed lines indicate constant Jeans mass for atomic gas (Equation 3). (*b*, *ii*) The main heating and cooling processes relevant for primordial gas. Cooling rates associated with molecular hydrogen are depicted with solid lines, other cooling processes with dashed lines, and the two main heating mechanisms with dotted lines. Data for panel *b*, subpanel *ii*, are provided by Kazuyuki Omukai and adapted with permission from ApJ. Abbreviations: CIE, collision-induced emission; ro-vib, rotation–vibration.

As a result, all atomic hydrogen is converted into H_2 once particle densities of $n \approx 10^{11}\ \mathrm{cm}^{-3}$ are reached. However, as collapse proceeds and the temperature exceeds values of ~2,000 K at densities of above $10^{13}\ \mathrm{cm}^{-3}$, the H_2 molecules become collisionally dissociated and the gas eventually turns atomic again.

With the appropriate chemical network and the corresponding heating/cooling functions included in cosmic structure formation calculations, we can follow the initial collapse phase from the cosmic mean up to the formation of the first hydrostatic core in the halo center. An example of this behavior is provided in **Figure 3*a***. The data are taken from a resimulation of a halo studied by Schauer et al. (2021) at a time just before the formation of the first protostar. We show the effective equation of state, i.e., the relation between temperature T and number density n of hydrogen atoms (**Figure 3*a***, subpanel *i*) as well as the corresponding fraction of molecular hydrogen (**Figure 3*a***, subpanel *ii*). The labels in the $\log_{10}T - \log_{10}n$ plot indicate key phases of the initial collapse: (A) As the gas begins to flow into the potential well of the dark matter halo, it is compressionally heated to the virial temperature of the system. At the same time the H_2 fraction

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increases from $<10^{-5}$ to $\sim 10^{-3}$, which is sufficient for the gas to go into a runaway cooling phase (B), and brings the gas down to the minimum temperature of $\sim$200 K (C). As more gas flows into the halo center, eventually the potential becomes dominated by the gas rather than by dark matter. The contraction proceeds, and the heat provided by *PdV* work begins to dominate again over cooling. As a consequence, the gas temperature rises to about 1,000 K at $n \approx 10^9$ cm^{-3} (D). At this stage, three-body H_2 formation (Equation 16) becomes important, and the gas quickly becomes fully molecular (E). Below densities $n \approx 10^{13}$ cm^{-3}, the main cooling process is the (forbidden) ro-vibrational line emission from H_2. However, at larger n, two additional cooling processes become important (F). The first is collision-induced emission (CIE), which occurs when two molecules come close to each other. Van der Waals forces can then induce a temporary dipole that allows for efficient dipole emission during the interaction time interval (Omukai & Nishi 1998, Ripamonti & Abel 2004). The second one is associated with the collisional dissociation of H_2, which sets in at temperatures around 2,000 K and quickly dominates the overall cooling behavior. As the density increases further (G), more and more H_2 molecules are destroyed. This also implies that more hydrogen atoms become available again for H_2 formation through the three-body process, and the associated energy release starts to dominate the overall heating rate over *PdV* work. This continues until the molecular gas is largely depleted, and the collapse becomes almost adiabatic at $n \approx 10^{20}$ cm^{-3}.

For comparison, we provide in **Figure 3*b*** (subpanel *i*) the relation between $\log_{10} T$ and $\log_{10} n$ from a one-zone astrochemical model describing the time-evolution of the central region during the initial collapse (see, e.g., Omukai & Nishi 1998). It essentially depicts the effective equation of state. We see that the polytropic index $\gamma_{\rm eff}$ varies considerably across the different regimes introduced above. Data are taken from Omukai et al. (2005, 2010) and describe the thermodynamic response of gas across a wide range of metallicities. We start with a purely primordial composition ($Z = 0$) and show the results for $\log_{10} Z/Z_\odot = -6, -5, \ldots$ up to the solar value ($Z = Z_\odot$). Lines of constant Jeans mass are also indicated.

For a more quantitative assessment of the most relevant heating and cooling processes during the initial collapse phase of gas in primordial halos, we also plot the corresponding rates as a function of the hydrogen number density, using data taken from the one-zone model of Glover & Savin (2009). Here, solid lines depict cooling processes associated with molecular hydrogen: ro-vibrational line emission from H_2 and HD, CIE of H_2 at high densities, and collisional dissociation of H_2. The cooling processes associated with other species are less relevant. Using dashed lines, we list cooling from H_3^+, LiH, Compton scattering, and H. In the absence of stellar feedback only two heating processes are important (*dotted lines*): *PdV* work dominates at densities below $n \approx 10^{13}$ cm^{-3}, and the latent heat released by H_2 formation is relevant at higher densities.

We note that over the roughly 18 orders of magnitude in density covered in **Figure 3**, the temperature of the primordial gas only varies by a factor of 25 or so. Overall the gas roughly exhibits an effective index of $\gamma_{\rm eff} \approx 1.08$, which is close to the isothermal value of $\gamma_{\rm eff} = 1$. This has important consequences for the level of fragmentation in primordial gas, and it is essential to understand the dynamical evolution of the accretion disk that inevitably builds up around the central object (see Section 3.2.2). We also note that **Figure 3** only considers compressional and chemical heating. The situation may change if radiative feedback from newly formed stars (Section 4.2) or the possible energy input from turbulent dissipation or from dark matter annihilation is considered (Section 3.3.2).

3.2.2. Disk formation and fragmentation. The first 3D simulations able to follow the initial cooling and collapse of the gas in high-redshift minihalos became available around the year 2000 (e.g., Bromm et al. 1999, 2002; Abel et al. 2000, 2002; Nakamura & Umemura 2001; Yoshida et al.

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2003). Although these numerical simulations were fully three-dimensional, the halos considered were relatively round and, thus, assuming spherical symmetry was a very good approximation during the early stages of collapse (see also Yoshida et al. 2006, 2008). These early calculations typically stopped when the object in the center reached hydrogen number densities of $n \approx 10^{16}$ cm^{-3}, above which the computational time step became prohibitively small. At this time, the hydrostatic object in the center had a mass of only $\sim 10^{-3}$ $M_\odot$. At such an early time, the material that made it to the center carried very little angular momentum, and this object was surrounded by only a small disk-like structure that was more strongly supported by pressure than by rotation. The authors of these studies suggested that this should be true for the entire protostellar accretion history and, hence, argued that all the inflowing mass would end up in one single high-mass star (see also Tan & McKee 2004). Clearly this supposition needed to be tested, in particular, because the statement that primordial stars only form in isolation is in tension with present-day star formation, in which fragmentation is ubiquitous and massive stars are typically found in clusters and aggregates (Lada & Lada 2003).

Figure 4*a*, adopted from a high-resolution simulation published by Greif et al. (2011a) ten years later, gives a visual impression of three star-forming halos at various spatial scales, eventually zooming in on the immediate physical environment where Pop III stars build up. The gas virializes on a scale of 5 kpc (comoving), followed by the runaway collapse of gas in the central 10 pc, where it becomes self-gravitating and decouples from the dark matter. In the final stages

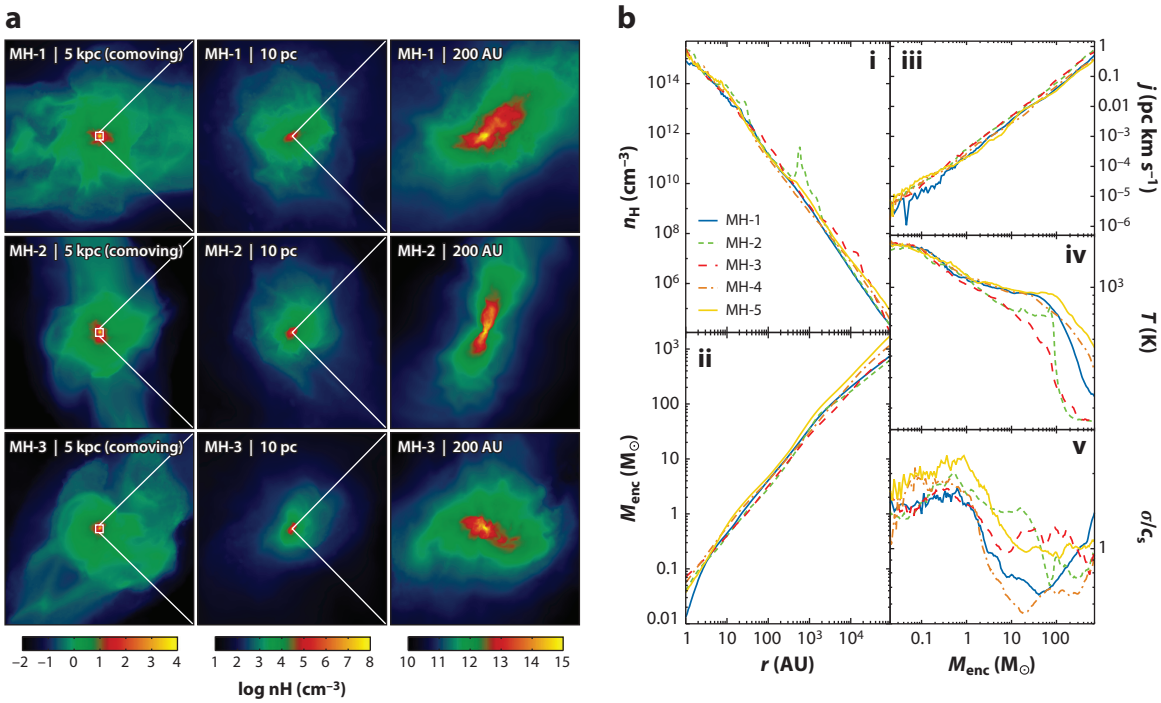

Figure 4

Structural and dynamical properties of three star-forming halos. (*a*) A sequence of zoom-ins focusing on the density-squared weighted hydrogen number density projected along the line of sight. (*b*) Cumulative mass as a function of radius (*i*) and corresponding number density profile (*ii*) in the inner regions of the halos (plus two additional systems not shown here) shortly before the onset of star formation. Specific angular momentum (*iii*), gas temperature (*iv*), and turbulent velocity dispersion (*v*) are plotted as functions of enclosed mass. Figure adapted from Greif et al. (2011a) with permission from ApJ.

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of the collapse, a fully molecular core forms on scales of ∼200 AU. **Figure 4*b*** provides key structural and dynamical parameters, including cumulative mass and the corresponding radial density profile to the left, as well as specific angular momentum, gas temperature, and turbulent velocity dispersion as a function of enclosed mass to the right. Although the morphological parameters of the halos are very similar on large scales, they differ on small scales. They also have quite distinct dynamical properties. The collapse induces turbulence in the infalling gas, which leads to stochastic variations in the structural appearance of the central regions, demonstrating that small differences in the initial fluctuation spectrum can amplify during the nonlinear collapse phase and lead to very different evolutionary pathways. For example, HD cooling became important in only two of the five halos modeled by Greif et al. (2011a), leading to lower temperatures in these halos and a different fragmentation behavior. This suggests that, similar to present-day star formation (McKee & Ostriker 2007, Klessen & Glover 2016), stellar birth in the early Universe is also subject to large statistical variations with the outcome sensitively depending on the details of the initial conditions and environmental parameters as well as on various complex nonlinear feedback loops.

Current simulations of Pop III star formation acknowledge this complexity and attempt to follow the formation and dynamical evolution of the accretion disk that builds up around the central object and ideally covers the entire accretion history, until stellar feedback removes the remaining gas and the process of stellar birth is completed. These studies demonstrate that primordial accretion disks are highly prone to fragmentation. They suggest that the standard pathway of Pop III star formation leads to a stellar cluster with a wide range of masses rather than the buildup of one single high-mass object. This is illustrated in **Figure 5**, taken from Clark et al. (2011b), showing the early evolution of a typical Pop III accretion disk. It fragments and builds up a multiple system of four protostars within only about one hundred years after the formation of the first object.

Disk fragmentation across a wide range of spatial and temporal scales is reported in essentially all current studies (e.g., by Machida et al. 2008a,b; Turk et al. 2009; Clark et al. 2011b;

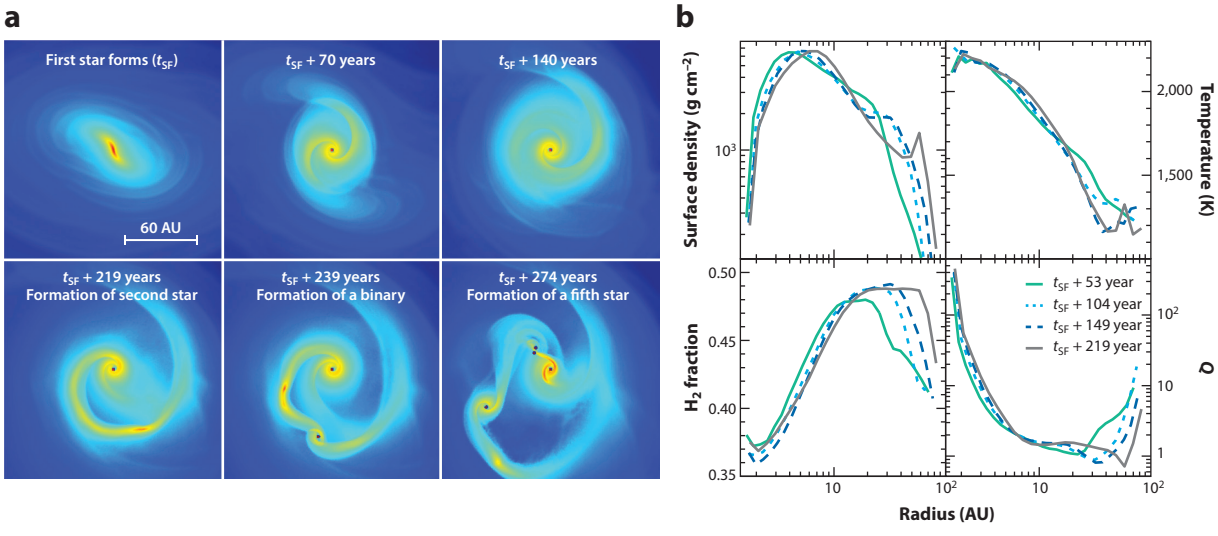

Figure 5

Formation and evolution of the disk around a Population III protostar. Panel *a* gives a visual impression of the buildup of a two-arm spiral structure caused by gravitational instability that subsequently fragments into four protostars within 240 years. Panel *b* shows the radial surface density and temperature profile at different times. It also demonstrates that the disk material is almost fully molecular ($n_{H_2}/n = 0.5$) and Toomre unstable ($Q < 1$). Figure adapted from Clark et al. (2011b) with permission of Science.

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Greif et al. 2011b, 2012; Smith et al. 2011, 2012; Dopcke et al. 2013; Stacy & Bromm 2013; Susa 2013, 2019; Vorobyov et al. 2013; Hirano et al. 2014; Susa et al. 2014; Hosokawa et al. 2016; Stacy et al. 2016; Takahashi & Omukai 2017; Hirano & Bromm 2018; Sugimura et al. 2020; Wollenberg et al. 2020; Sharda et al. 2021; Chiaki & Yoshida 2022; Jaura et al. 2022; and Prole et al. 2022a, which is by no means an exhaustive list). They predict the formation of a cluster of multiple Pop III (proto)stars, which grow in mass at rates of $\dot{M} \approx 10^{-3}\,\mathrm{M}_\odot\,\mathrm{year}^{-1}$ with possible brief periods of accretion with rates as high as a few times $10^{-2}\,\mathrm{M}_\odot\,\mathrm{year}^{-1}$. The reason for the high susceptibility of primordial accretion disks to fragmentation is always the same: For typical halo conditions in the primordial Universe, the mass load onto the disk from the infalling envelope exceeds its capability to transport material inward by gravitational or magnetoviscous torques. As a consequence, massive spiral arms build up and speed up the inward transport. Often this is not enough, and the arms become nonlinear and interact with each other, leading to runaway collapse in the interaction regions, as is clearly visible in **Figure 5**. The disk is fragmenting and forms new protostars. As this happens, a smaller accretion disk forms around the new object, which by itself may become unstable and fragment to form additional protostars. For very quiescent initial conditions with low levels of turbulence, this secondary process may be the prevalent form of fragmentation (Susa 2019). As the global accretion disk gains more mass and grows in size by accretion of higher angular momentum material, the Toomre unstable region moves further out. Consequently, fragmentation and formation of new protostars occurs at larger and larger radii as the evolution progresses.

Hill volume: region of influence of a smaller body in the face of gravitational perturbations from a more massive body

3.2.3. Population III initial mass function and multiplicity. As argued in the previous section, it is almost impossible to avoid the fragmentation of primordial accretion disks. The key question then is, what happens to the fragments during their subsequent dynamical evolution? There are three possible outcomes: (*a*) A fragment grows in mass and survives to become a proper star (or substellar object), or (*b*) it dies by getting swallowed by the central object, or (*c*) it merges with another fragment, in which case the same question applies to the new object. A related question is, if the fragment survives, what sets its final stellar mass? Again, there are three answers possible. (*a*) The system simply runs out of mass, leaving behind relatively massive stars with interesting implications for their detectability at high redshifts (Section 5.1). (*b*) Stellar feedback removes material from the system and terminates any subsequent accretion. This is the most complex case and can lead to a wide range of different results, with the currently available simulations being highly inconclusive, as discussed in Section 4.2.1. (*c*) The fragment gets ejected from the disk by dynamical interactions and its growth is stopped prematurely. This typically leads to low-mass objects, some of which could potentially have survived until the present day and might be detectable in stellar archeological surveys (see the discussion in Section 5.2).

Let us consider these processes in more detail. As material flows through the disk toward the center, it first encounters the Hill volume of protostars that lie further out, and so it preferentially gets swallowed by these objects rather than by those close to the center. This reduction of the mass growth rate is more pronounced at smaller radii and can lead to the complete starvation of the primary object in the very center (Peters et al. 2010, Girichidis et al. 2011). In simple binary systems, this process results in the two constituent stars having roughly equal masses (see also Kratter & Matzner 2006). In the more complex situation of Pop III star formation, with ejections, mergers, and absorption by the central object, all of which are unpredictable and highly stochastic processes, we expect a wide spectrum of stellar masses.

When considering the question of what fraction of fragments merge or become swallowed by the central object and how many survive, various studies (e.g., Greif et al. 2012, Smith et al. 2012, Stacy & Bromm 2013) indicate that roughly two-thirds of the fragments quickly disappear

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again and about one-third remain. These numbers should be taken with caution, as none of the simulations covers the entire duration of disk evolution, with the highest-resolution models only able to cover the least amount of time. It therefore could be that protostars that are counted as ejected will eventually fall back again and become accreted then. Also those remaining in the disk may still merge at a later stage. In addition, the result also depends on how mergers are numerically implemented (Wollenberg et al. 2020). Still, as long as fragmentation in the disk continues and the appearance and disappearance of new protostars is an ongoing process, adopting a ratio of roughly 1:2 of survivors and mergers is a good estimate. It has also been suggested that the number $N_{\rm frag}$ of fragments, survivors as well as mergers, increases with time t as $N_{\rm frag} \propto t^{0.3}$ Susa (2019).

Initial mass function (IMF): the stellar IMF describes the statistical distribution of stellar masses at birth

Even if Pop III stars form in multiple systems across a wide range of masses, we can nevertheless look for the most bound pairs and investigate their properties. Doing so, Stacy & Bromm (2013) and Stacy et al. (2016) report a binary fraction of ~35%, with semimajor axes as large as 3,000 AU and with a wide distribution of orbital periods, ranging from ~30 to 30,000 years, with the smallest period being determined by the minimum numerical resolution achieved. They also find that the distribution of mass ratios is relatively flat (as also suggested by Susa 2019). With these data, it is possible to build large ensembles of Pop III binary systems for long-term integration in dedicated *N*-body simulations or semianalytic models (e.g., Liu et al. 2021, Santoliquido et al. 2021).

Concerning the resulting distribution of stellar masses, the initial mass function (IMF) of Pop III stars, the ubiquity of fragmentation and the stochasticity of the processes involved lead to a wide range of stellar masses, with all current models suggesting values from the substellar regime up to several hundred solar masses. As mass is the most important parameter in stellar evolution theory (Kippenhahn et al. 2012), predicting the Pop III IMF is a central aspect of many studies. In **Figure 6**, we present the results of some of these simulations to give an indication of the diversity of the current state of the art. We select three high-resolution models without stellar feedback: (1) model $\rho_{\rm sink} = 10^{-6}\,{\rm g\,cm^{-3}}$ from Prole et al. (2022a), (2) model $\alpha005\beta001$ from Wollenberg et al. (2020), and (3) the combined model from Stacy & Bromm (2013). Models 1 and 2 reach very high spatial resolution but only cover a small fraction of the total accretion time in the halo. Model 3 is not so well resolved but covers more time. The effect of resolution is noticeable in the resulting mass spectrum. We also chose three simulations with radiative feedback from newly formed stars: (4) T_RTP from Jaura et al. (2022), (5) III.1 from Hirano et al. (2015), and (6) the combined models from Hirano et al. (2014). Model 4 has similar resolution and time coverage as the first two examples (models 1 and 2) without feedback, and it predicts a similar mass spectrum. Models 5 and 6 have somewhat lower resolution but cover considerably more time until the protostellar accretion has stopped. Whereas models 1–4 are fully three-dimensional simulations, models 5 and 6 have been performed in two dimensions assuming axisymmetry. These calculations predict considerably larger masses. Altogether, the reported stellar mass spectra vary enormously, which is likely a consequence in part of the different resolutions and periods simulated in the different calculations. However, it is a common feature that they all are approximately logarithmically flat. This is in stark contrast to the present-day IMF, which shows a clear peak around 0.2–0.3 $M_\odot$ followed by a steep power-law falloff (Kroupa 2002, Chabrier 2003). For this reason, the IMF of primordial stars is often called top-heavy in comparison.

We conclude that our understanding of the IMF of primordial stars is still quite limited. The existing models all have specific shortcomings, with none being able to reach sufficiently high resolution, cover enough time, and properly include all relevant physical processes. As indicated in **Figure 6**, we find that models that include stellar feedback on average predict larger stellar masses than those without. In addition, the outcome strongly depends on the spatial resolution

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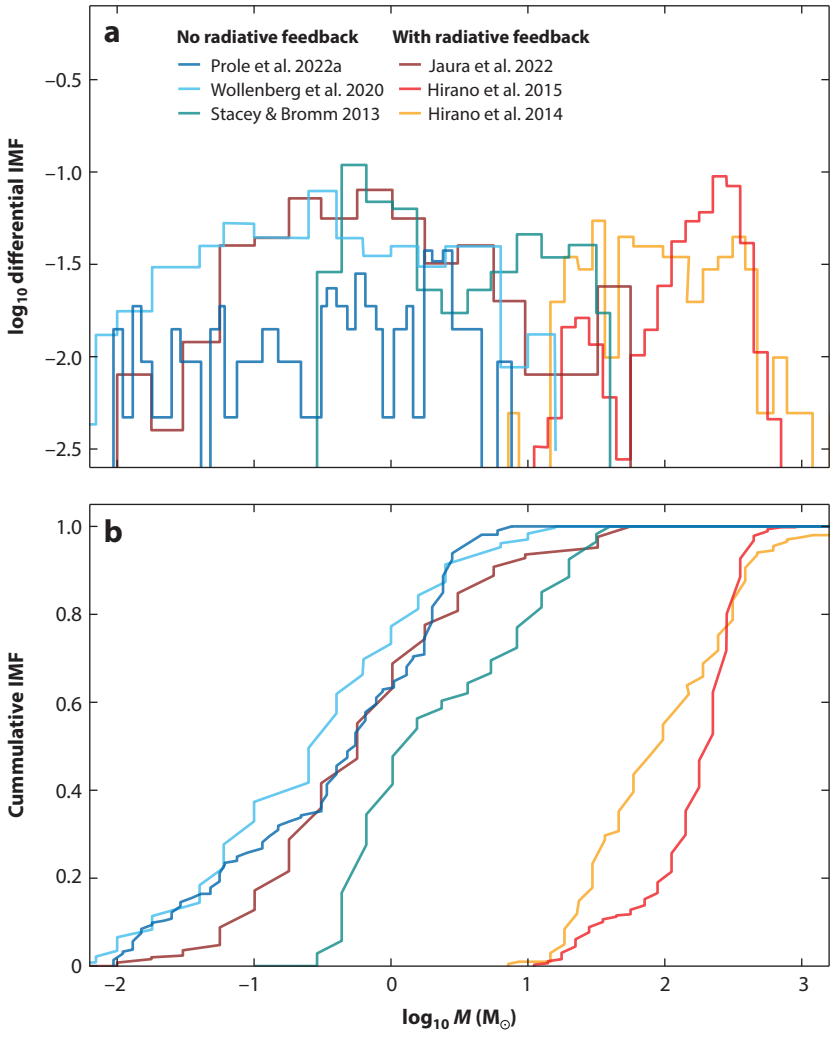

Figure 6

(*a*) Normalized differential and (*b*) cumulative stellar mass distribution obtained from six different studies of Population III star formation. To represent the current state of the art, we select three high-resolution models without stellar feedback (Prole et al. 2022a, model with $\rho_{\rm sink} = 10^{-6}\,{\rm g\,cm^{-3}}$; Wollenberg et al. 2020, model $\alpha005\beta001$; Stacy & Bromm 2013, all models) and three simulations with radiative feedback included (Jaura et al. 2022, model T_RTP; Hirano et al. 2015, model III.1; Hirano et al. 2014, all models). Despite the large variation in range, the predicted IMFs are rather flat compared to the present-day IMF. Abbreviation: IMF, initial mass function.

achieved, the time span covered, and the details of the numerical implementation. High-resolution simulations tend to yield smaller stellar masses, because they are better able to resolve disk fragmentation (see, e.g., the discussion in Stacy et al. 2016 or Prole et al. 2022a), but they typically only cover a small fraction of the full accretion history of the halo. This raises the question of whether the remaining gas reservoir is mostly used to form new (low-mass) objects or whether it is largely consumed by existing protostars, which then continue to grow in mass. Similarly,

simulations covering a large fraction of the halo collapse timescale do not resolve the inner accretion disk well and, therefore, exhibit lower levels of fragmentation. Hence, they are biased toward higher-mass stars. Furthermore, two-dimensional simulations tend to report less fragmentation than full three-dimensional ones. Fragmentation is also influenced by additional physical processes, which we discuss in Section 3.3 below.

UV photons: UV photons in the energy range of $h\nu \geq 13.6$ eV can photoionize H

LW photons: Lyman–Werner photons in the energy range of $11.2\ \text{eV} \leq h\nu < 13.6$ eV can photodissociate H_2 molecules

3.2.4. Intermediate conclusions. Altogether, we can draw the following conclusions from the discussion so far: First, once a halo decouples from the cosmic expansion, the question of whether or not it forms stars depends strongly on its ability to cool. Second, as gas flows in and assembles in an accretion disk, it is highly susceptible to fragmentation. Consequently, the formation of binaries or higher-order multiple stellar systems is the norm of Pop III star formation rather than the exception. Third, when focusing on binary stars and the most bound pairs in a hierarchical system, we expect a wide range of separations, a flat mass ratio distribution, and a roughly thermal spread of orbital eccentricities. Fourth, the resulting stellar mass spectrum is very wide, ranging from low-mass stars, which potentially could have survived until the present day with interesting implications for stellar archaeological surveys (Section 5.2.1), to very massive stars, which produce copious amounts of UV photons and LW photons in the early Universe. Overall the resulting mass spectrum is likely to be rather flat (in the logarithm of mass) and therefore top-heavy compared to the present-day values. Fifth, developing a better understanding the impact of stellar feedback is one of the key challenges of current research into Pop III star formation. Specifically, radiative feedback is very likely to reduce the number of stars within a star-forming halo and lead to larger stellar masses (see also Section 4.2). However, it can also potentially increase the level of (large-scale) fragmentation in externally irradiated halos (Section 4.3).

3.3. Alternative Population III Star-Formation Pathways

Here, we touch upon alternative Pop III star-formation scenarios and speculate about the impact of physical processes that are often neglected in numerical and analytical models. We group our discussion into processes that can promote fragmentation in the primordial gas and those that can reduce the level of fragmentation. We defer the discussion of the most extreme physical conditions, which might lead to the formation of SMSs and the seeds of SMBHs in the Universe, to Section 3.4, and we note that a detailed analysis of the impact of stellar feedback is the focus of Section 4. We also note in passing that adopting different cosmological models compared to standard ΛCDM (Section 2.1) can further modify the picture. It is well understood, for example, that in a warm dark matter scenario star formation happens later and in larger halos (Dayal et al. 2015, Maio & Viel 2015, Magg et al. 2016, Mocz et al. 2020). A similar result has been reported for fuzzy dark matter models (e.g., Mocz et al. 2019). However, it remains to be seen whether these global changes actually influence the properties of the individual stars that form, and we do not follow up further on this topic here.

3.3.1. Enhancing fragmentation. Several numerical studies indicate that fragmentation also occurs on larger scales in the halo, on scales of the star-forming cloud as a whole (Turk et al. 2009, Stacy et al. 2010, Clark et al. 2011b) rather than just at the level of the central disk, as discussed in Section 3.2.2. It is typically associated with higher levels of turbulence in the halo gas. The initial turbulent flows that are always present with subsonic or transonic velocities get amplified during gravitational collapse and induce density fluctuations that can go into runaway growth in their own right. This is very similar to the turbulence-driven mode of star formation that is dominant at the present day (Elmegreen & Scalo 2004, Mac Low & Klessen 2004, Scalo & Elmegreen 2004, McKee & Ostriker 2007, Krumholz 2015, Klessen & Glover 2016). There are two systems

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for which this effect is thought to be particularly important: atomic cooling halos and halos that are subject to large relative streaming velocities between baryons and dark matter.

3.3.1.1. External irradiation and atomic cooling halos. Atomic cooling halos are halos with virial temperatures high enough to allow cooling by atomic hydrogen, i.e., $T_{\rm vir} > 8{,}000$ K. In general, we expect a Pop III star to form in a halo once its virial temperature exceeds a few thousand Kelvins (see Section 3.1) and, hence, that most atomic cooling halos will be associated with metal-enriched gas. However, in the presence of a sufficiently strong LW radiation field, H_2 cooling and star formation can be suppressed in halos with $T_{\rm vir} < 8{,}000$ K (see, e.g., Agarwal et al. 2019, Oh & Haiman 2002, and also Section 4.3.2). In that case, cooling only gets underway once the halo has grown to the point that $T_{\rm vir} > 8{,}000$ K. Then the gas starts to cool quasi-isothermally via Lyman-α emission. During this initial period of cooling, the high gas temperature keeps the critical mass for runaway collapse high (see Section 3.1). This implies that once collapse becomes possible, the associated infall velocities and rates, which we can infer from Equation 4, are larger than the ones discussed in Section 3.2.1. The high gas temperature also keeps the gas partially ionized, enabling it to form H_2 rapidly once it becomes dense enough to shield itself from the external LW radiation field, resulting in a transition to efficient H_2 cooling and a rapid temperature drop (Oh & Haiman 2002). The combination of large velocities and low temperatures means that the turbulence associated with the inflow motion is likely to become trans- and supersonic (Wise & Abel 2007a, Greif et al. 2008, Wise et al. 2008). The flow is characterized by cold streams that bring dense material rapidly to the center, where it can efficiently fragment and form stars. It has been suggested that this so-called Pop III.2 mode of primordial star formation leads to a different IMF (McKee & Tan 2008, Clark et al. 2011b, Maio et al. 2011, Stacy et al. 2011), but overall the results are not fully conclusive.

3.3.1.2. Streaming velocity between baryons and dark matter. High levels of turbulence are also expected in regions of large relative streaming velocity between baryons and dark matter (Section 3.1.2). Simulations that include this effect suggest that it reduces the gas overdensity in low-mass halos, delays the onset of cooling, and leads to a larger critical mass for collapse to set in (see, e.g., Greif et al. 2011b; Maio et al. 2011; Stacy et al. 2011; Naoz et al. 2012, 2013; O'Leary & McQuinn 2012; Latif et al. 2014a; Schauer et al. 2017b, 2020, 2021). It may also have substantial impact on the resulting 21-cm emission (Fialkov et al. 2012, McQuinn & O'Leary 2012, Visbal et al. 2012; see also Section 5.1.3). Gas in halos that are subject to large streaming velocities is also more turbulent than gas in more quiescent systems, and so we expect more fragmentation and a bias toward smaller stellar masses (Clark et al. 2008). However, this process has not yet been modeled with sufficient resolution, and so no reliable predictions of its impact on the IMF exist.

3.3.2. Reducing fragmentation. Besides stellar feedback, which we address in detail in Section 4, two additional physical processes have been suggested to reduce the level of fragmentation and, thus, to influence stellar IMF and multiplicity. These are magnetic fields and dark matter annihilation, both of which we discuss below.

3.3.2.1. Magnetic fields. The presence of dynamically important magnetic fields could significantly alter the picture presented so far. We know that the current Universe is highly magnetized on all scales (Beck et al. 1996) and that this influences the birth of stars and the evolution of the interstellar medium (ISM).[2] The properties of the magnetic fields observed today are well explained by a combination of small-scale and large-scale dynamo processes (Brandenburg & Subramanian

[2]For the extreme viewpoint of magnetically mediated star formation, see Shu et al. (1987).

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2005). In contrast, our knowledge of magnetic fields at high redshifts is very sparse. Theoretical models predict that magnetic fields could be produced in various ways, for example, via the Biermann battery (Biermann 1950), the Weibel instability (Medvedev et al. 2004, Lazar et al. 2009), or thermal plasma fluctuations (Schlickeiser & Shukla 2003). Other theories place their origin in cosmological phase transitions or during inflation (Sigl et al. 1997, Grasso & Rubinstein 2001, Banerjee & Jedamzik 2003, Widrow et al. 2012). The resulting fields are thought to be orders of magnitudes too weak to have any dynamical impact, so magnetohydrodynamic effects have often been neglected in numerical simulations of primordial star formation (however, see the analytic models of Pudritz & Silk 1989, Tan & McKee 2004, Silk & Langer 2006).

This situation has changed with the realization that the small-scale turbulent dynamo can efficiently amplify even extremely small primordial seed fields to the saturation level (Kulsrud et al. 1997), and that this process is very fast, acting on timescales much shorter than the free-fall time. An analytic treatment is possible in terms of the Kazantsev model (Kazantsev 1968; Subramanian 1998; Schober et al. 2012a,b). This describes how the twisting, stretching, and folding of field lines in turbulent magnetized flows leads to exponential growth of the field. The amplification timescale is comparable with the eddy-turnover time on the viscous or resistive length scale, depending on the value of the magnetic Prandtl number (the ratio of the kinematic viscosity to the magnetic diffusivity). Once backreactions become important, the growth rate slows down and saturation is reached within a few large-scale eddy-turnover times (Schekochihin et al. 2004, Schober et al. 2015, Liu et al. 2022). Depending on the properties of the turbulent flow, the magnetic energy density at saturation is thought to lie between 1% and a few 10% of the kinetic energy density with a field topology that is highly tangled (Federrath et al. 2011, Seta & Federrath 2021).

Magnetic fields with that strength can strongly affect the evolution of protostellar accretion disks. If the field has a strong poloidal component, it can efficiently remove angular momentum from the star-forming gas and reduce its level of fragmentation (Machida et al. 2008a,b,c; Bovino et al. 2013; Latif et al. 2013, 2014b; Machida & Doi 2013; Hirano & Machida 2022; Saad et al. 2022), which influences the resulting IMF (Turk et al. 2011, Peters et al. 2014, Sharda et al. 2021). A highly ordered small-scale field can also drive protostellar jets and outflows (Machida et al. 2006, Sadanari et al. 2021). Finally, the presence of a dynamically significant magnetic field can also change the rotational properties of Pop III stars (Machida et al. 2007, Stacy et al. 2013), which in turn has consequences for their expected lifetimes and overall luminosity (see Section 4.2). One important caveat, however, is that many studies of the impact of the magnetic field assume that the field is highly ordered on small scales. This is not true initially for a field amplified by the small-scale turbulent dynamo, and although we expect the field to become more ordered over time, it remains unclear how rapidly this occurs, with recent simulations yielding contradictory results (Sharda et al. 2021, Prole et al. 2022b, Saad et al. 2022, Stacy et al. 2022).

Altogether, we expect Pop III clusters to have fewer members with somewhat higher masses than predicted by purely hydrodynamic simulations, such as is discussed in Section 3.2. However, the details depend very much on the adopted field topology, with initially smooth and homogeneous fields leading to more magnetic braking then highly tangled turbulent fields (e.g., Kuruwita & Federrath 2019, Seifried et al. 2012) as expected to result from the small-scale turbulent dynamo at work (e.g., Seta & Federrath 2020). Thus, the questions of how magnetic fields influence the overall star-formation process in primordial gas and how they affect the resulting IMF remain subject to very active research.

3.3.2.2. Dark matter annihilation. Despite its importance for cosmic evolution and structure formation, the true physical nature of dark matter is still unknown. Many models introduce a new class of weakly interacting massive particles (WIMPs), as they naturally occur in supersymmetry

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theories (e.g., Jungman et al. 1996). The lightest supersymmetric particle is expected to be stable and have properties consistent with the phenomenological requirements on dark matter (Bertone et al. 2005). If these particles are self-annihilating, they will act as an additional source of heating.

In most environments, the dark matter density is too low for this to be significant. However, this may be different in the very centers of star-forming halos in the early Universe. Here, the collapse of the baryons may lead to adiabatic contraction of the dark matter halo (Blumenthal et al. 1986), increasing its central density by several orders of magnitude. As the annihilation rate scales quadratically with density, the corresponding energy input and ionization rate may become large enough to influence gas dynamics. Spolyar et al. (2008) and Freese et al. (2009) suggest that this process may overcome the cooling provided by H_2, and they speculate that this could halt gravitational collapse and lead to the formation of so-called dark stars. If dark matter particles also scatter weakly on baryons, these dark stars would be stable for a long time without ever becoming dense or hot enough to initiate nuclear fusion. They would be much larger and more massive than normal Pop III stars, with sizes of a few astronomical units, lower surface temperatures, and higher luminosities (Freese et al. 2008, Iocco 2008, Iocco et al. 2008, Yoon et al. 2008, Hirano et al. 2011).

There are several problems with this scenario. First, it is not clear whether collapse stalls once the energy input from dark matter becomes comparable with the cooling rate. Ripamonti et al. (2010) argue that this is not the case because the larger heating rate catalyzes further formation of H_2 and is compensated by the corresponding larger cooling rate. In addition, chemical cooling due to H_2 collisional dissociation can help to balance the additional heating after only a moderate increase in the gas temperature (Smith et al. 2012). Second, the implicit assumption of perfect alignment between the dark matter cusp and gas collapse is most likely violated in realistic star-formation conditions. Three-dimensional simulations (Stacy et al. 2012, 2014) clearly demonstrate that the presence of nonaxisymmetric perturbations leads to a separation between the dark matter cusp and collapsing gas, rendering the annihilation energy input insignificant for dark star formation. However, it is still possible that dark matter annihilation influences the dynamics of the accretion disk in the halo center and that the energy input associated with this process leads to a suppression of disk fragmentation (Smith et al. 2012). We conclude that dark matter annihilation may be able to reduce the multiplicity of metal-free stars and increase their overall mass, but we also note that the existing studies are still premature, and it is too early for a reliable assessment of the impact of this process on the IMF of Pop III stars.

3.4. Supermassive Stars and Black Holes

The existence of extremely bright quasars at redshifts $z \gtrsim 7$ (see, e.g., Fan et al. 2006b, Wu et al. 2015, Bañados et al. 2018, Wang et al. 2021) implies the presence of SMBHs with masses of $M_\bullet \sim 10^9\,M_\odot$ and above. This finding is in tension with physical models in which the seeds of SMBHs start out light and then grow gradually by Eddington-limited accretion. The standard Pop III star-formation pathway discussed in Section 3.2 produces black holes at the end stage of stellar evolution with $M_\bullet < 10^2\,M_\odot$ (see also Section 4.3.4). These small seeds do not have enough time to grow to the observed large masses over the age of the Universe ($\sim$0.7 Gyr at $z = 7$, assuming standard ΛCDM cosmology). In addition, the assumption of persistent Eddington-limited accretion is itself questionable. Stellar feedback from Pop III stars is very efficient at removing gas from their birth sites, leaving little to be accreted by their black hole remnants, and the motion of these black holes within their parent minihalos can also be substantial, further reducing their accretion rate (Smith et al. 2018). Therefore, more extreme formation scenarios need to be considered. We briefly review a few of the most important models here and refer the reader to Woods et al. (2019), Latif & Schleicher (2019), or Inayoshi et al. (2020) for more comprehensive reviews.

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3.4.1. Need for high initial infall rates. The time taken for a black hole to grow via Eddington-limited accretion from an initial mass $M_{\bullet,i}$ to a final mass $M_{\bullet,f}$ is given by

$$\tau_\bullet = \frac{0.4}{f_{\rm Edd}} \frac{\epsilon}{1-\epsilon} \ln\left(\frac{M_{\bullet,f}}{M_{\bullet,i}}\right) \text{ Gyr}, \qquad 17.$$

where ϵ is the accretion efficiency, typically $\sim$0.1 for accretion through a thin disk (Shakura & Sunyaev 1973), and where $f_{\rm Edd}$ expresses the fraction of time for which the black hole is able to accrete at the full Eddington rate (for a comprehensive review, see Inayoshi et al. 2020). To build an SMBH of $10^9\,{\rm M}_\odot$ at $z = 7$ starting from an initial seed mass of $10^2\,{\rm M}_\odot$ requires $\tau_\bullet \approx 0.8$ Gyr, which is longer than the age of the Universe at this redshift, even if we make the unrealistic assumption that $f_{\rm Edd} = 1$ throughout. Reducing $f_{\rm Edd}$ makes this discrepancy even larger. In principle, accretion at a rate faster than the Eddington limit is possible if the accretion flow strongly deviates from spherical symmetry and material gets delivered to the black hole in a highly filamentary fashion, or alternatively, if the accretion disk is radiatively inefficient or the accreting envelope emits anisotropically (e.g., Mayer & Bonoli 2019). However, the inferred small quasar duty cycles would require rates considerably above the Eddington limit during the periods of accretion. This seems unlikely, and so the preferred scenario for the formation of SMBHs at high redshifts is to start out with more massive seeds, i.e., to increase $M_{\bullet,i}$ to a much larger value.

To build such a massive seed object requires very high accretion rates, which in turn requires extreme environmental conditions. Clearly, the halo in which this happens needs to be massive enough to contain a sufficient amount of gas. To form a seed with a mass of $\sim 10^5\,{\rm M}_\odot$, we therefore need a halo with a mass $M \gg 10^7\,{\rm M}_\odot$, unless we assume that an improbably large fraction of the gas ends up in the seed black hole. This is much larger than the critical halo mass required for Pop III star formation and, hence, points to scenarios in which collapse and star formation are delayed in some fashion. Inayoshi et al. (2020) review the different models that have been suggested to explain this, ranging from irradiation of the halo by a high flux of LW photons (e.g., Agarwal et al. 2012) to dynamical heating of the gas by repeated major mergers (e.g., Mayer & Bonoli 2019).

Once we have physical conditions that allow a large amount of gas to flow into the center of an appropriately massive halo at rates of $\dot{M} \sim 0.1\,{\rm M}_\odot\,{\rm year}^{-1}$ or higher, the next question is whether this gas feeds the growth of a single object or whether the gas fragments and forms multiple objects. We briefly consider both possibilities below.

3.4.2. Supermassive stars. If we assume that only a single object forms—a scenario often referred to as direct collapse—then we can ask about the properties of this object and about the maximum permitted mass. These questions can be addressed with detailed stellar evolution calculations that include a treatment of accretion (e.g., Stahler et al. 1986a, Behrend & Maeder 2001, Omukai & Palla 2003, Hosokawa & Omukai 2009, Haemmerlé et al. 2016). They show that stable SMSs are possible and can reach maximum masses of up to several $10^5\,{\rm M}_\odot$ (Umeda et al. 2016, Woods et al. 2017, Haemmerlé et al. 2018), but always stay below $10^6\,{\rm M}_\odot$ for the accretion rates reachable in atomic cooling halos (Haemmerlé et al. 2019; Haemmerlé 2020, 2021a,b). This value is determined by the onset of the general relativistic (GR) instability, which drives the object into collapse to form a black hole with the same mass (see Appenzeller & Fricke 1972a,b or Fuller et al. 1986 for early models, or see Haemmerlé et al. 2021 for a more recent account). In **Figure 7**, we report estimates of the maximum mass from different sets of stellar structure and evolution calculations for different accretion rates and from simplified analytic models. The stellar structure calculations indicate that primordial SMSs evolve as red supergiant protostars (Hosokawa et al. 2012a, 2013; Haemmerlé et al. 2018), with extended radii that follow the relation given by Equation 21 and surface temperatures of only $\sim$5,000 K. Their internal structure consists of a convective core, a radiative zone containing most of the stellar mass, and a convective envelope that

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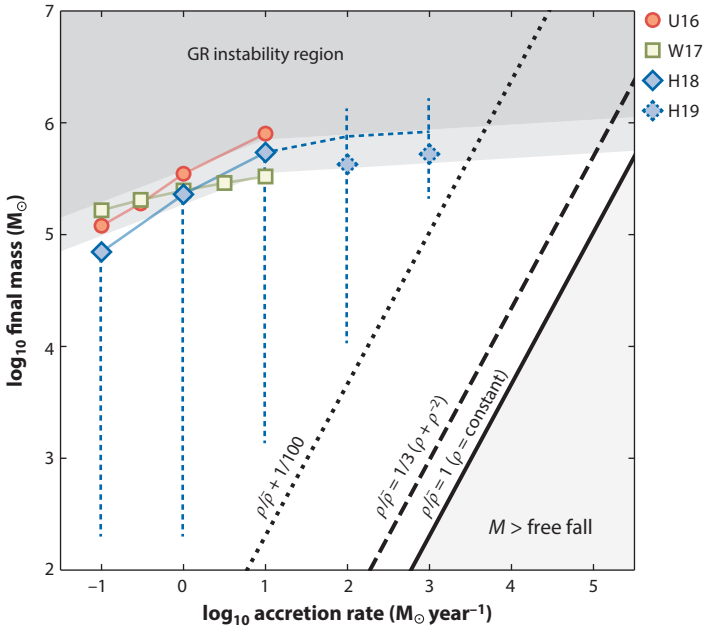

Figure 7

Maximum mass reachable by SMSs accreting at different rates until the onset of the GR instability triggers collapse to a black hole of the same mass. The colored symbols depict the final masses in the studies of Umeda et al. (2016), Woods et al. (2017), and Haemmerlé et al. (2018) for accretion rates up to $\dot{M} = 10\,M_\odot$ year^{-1}. We also report results from Haemmerlé et al. (2019) for $100\,M_\odot$ year^{-1} and $1{,}000\,M_\odot$ year^{-1}, but note that the calculations terminated before the GR instability sets in, and so the expected final masses are higher than shown here, as indicated by the tentative extension of the accretion path (*vertical dotted lines*). We also indicate the region forbidden by the GR instability as derived from analytic estimates (see Haemmerlé 2020, 2021a). The tilted lines show the evolution in M versus $\dot{M}$ space for the free-fall collapse of gas spheres with different initial density profiles, characterized by the contrast $\rho/\bar{\rho}$ between the density of the infalling gas and the mean density of the accreting SMS, following the mass–radius relation for high accretion rates (Equation 21). The figure is inspired by Haemmerlé et al. (2021). Abbreviations: GR, general relativistic; SMS, supermassive star.

covers a dominant fraction of the photospheric radius. To a good approximation, these structures can be described as hylotropes (Begelman et al. 2008, Begelman 2010), in particular for accretion rates $\gtrsim 10\,M_\odot$ year^{-1}. Although very luminous, their low surface temperatures do not allow them to emit large amounts of ionizing photons. Consequently, they are not able to create extended HII regions (Section 4.2.1), which might limit the overall mass growth or affect star formation in neighboring halos (Section 4.3.1).

3.4.3. Dense clusters. As argued in Section 3.2.2, it is difficult to prevent primordial gas from fragmenting. This is well established in the standard Pop III formation scenario, but it also appears to hold in highly irradiated atomic cooling halos in which H_2 is suppressed (e.g., Agarwal et al. 2012, Sugimura et al. 2014, Agarwal & Khochfar 2015, Latif et al. 2015, 2020, Regan & Downes 2018). It becomes even harder to prevent fragmentation in the presence of metals or dust, owing to the additional cooling channels that they provide (Omukai et al. 2008, Latif et al. 2016, Chon & Omukai 2020). Therefore, rather than forming a single star, it is plausible that the rapid accretion flows discussed here instead feed the growth of a dense cluster of objects, each accreting at its own pace. In this scenario, SMBH seeds result from runaway collisions between stars in this

HII region: bubbles of ionized gas surrounding high-mass stars or massive star clusters caused by radiative stellar feedback

dense environment (e.g., Portegies Zwart & McMillan 2000, 2002; Sesana et al. 2005; Devecchi & Volonteri 2009; Devecchi et al. 2012; Katz et al. 2015; Sakurai et al. 2017; Reinoso et al. 2018, 2020; Escala 2021; Vergara et al. 2021) or between their black hole remnants (Davies et al. 2011, Lupi et al. 2014, Antonini et al. 2019, Kroupa et al. 2020). Although many models have treated this as a purely stellar dynamical problem, the difficulty of disrupting these high accretion flows means that in practice one should simultaneously take gas dynamics and stellar dynamics into account. In this context, Davies et al. (2011) and Lupi et al. (2014) argue that the inflow of gas into an interacting cluster of stellar mass black holes is needed to steepen the gravitational potential and make mergers more likely than three-body ejections. Focusing on embedded star clusters, Boekholt et al. (2018) demonstrate that the combination of accretion and collisions can lead to masses up to $\sim 10^5\ M_\odot$. The semianalytic models of Tagawa et al. (2020) consider the growth of a supermassive object via stellar bombardment in the presence of gas. Other models study the impact of different accretion prescriptions (Das et al. 2021b), or of mass loss occurring in mergers (Alister Seguel et al. 2020) or associated with stellar winds (Das et al. 2021a). The impact of varying the metallicity of the gas has also been investigated (Chon & Omukai 2020, Schleicher et al. 2022).

The emerging picture is that fragmentation leads to the formation of a dense and deeply embedded cluster in which competitive accretion of individual cluster members in concert with frequent merger events results in the runaway growth of a small number of objects. Depending on the environmental conditions and on the detailed implementation of the physical processes considered, masses of order of $M_\bullet \sim 10^5\ M_\odot$ are easily within reach, and it is reasonable to speculate that the internal structure of these very massive runaway objects is similar to that of the SMSs discussed above. This is supported by calculations adopting highly time-varying accretion rates (Woods et al. 2021a,b) and by considering SMS mergers as particularly extreme forms of accretion spikes. They collapse into massive black holes once this runaway growth phase ends or once they reach the mass limit for the GR instability.

4. FEEDBACK FROM POPULATION III AND TRANSITION TO POPULATION II

The formation of Pop III stars is associated with a range of different feedback processes that affect the gas around them. Radiative and mechanical feedback in various forms influences the star-formation efficiency of the gas on both small and large scales, and chemical feedback, i.e., the enrichment of the Universe with metals from the first SNe, drives the transition from Pop III to Pop II star formation. In this section, we review the most important feedback processes. We begin with radiation (Section 4.1), and successively include other forms of stellar feedback, focusing first on their effects on gas close to the stars (Section 4.2) and later on their effects on much larger scales (Section 4.3). We also briefly review the physics of the Pop III to Pop II transition and the most important open questions associated with this (Section 4.4).

4.1. Radiation from Population III Stars

Here, we summarize the properties of the radiation produced by primordial stars. This sets the stage for the discussion of the impact of radiative feedback on the star-forming cloud itself and on neighboring halos.

4.1.1. Accretion luminosity. The first form of feedback to become important in a star-forming minihalo is the accretion luminosity generated by gas falling onto newly formed protostars. In principle, this represents a considerable reservoir of energy, owing to the high accretion rates

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typical in systems forming Pop III stars. A protostar with a mass M_* and radius R_* that is accreting gas at a rate $\dot{M}$ produces an accretion luminosity

$$L_{\rm acc} = \alpha \frac{GM_*\dot{M}}{R_*}, \qquad 18.$$

where α is a dimensionless efficiency factor that depends on the geometry of the accretion flow (Stacy et al. 2016), with high-resolution simulations (e.g., Wollenberg et al. 2020, Jaura et al. 2022) suggesting values of α close to unity. Accretion rates onto Pop III protostars can vary substantially from star to star, but at early times, values as high as 10^{-3}–10^{-2} $\mathrm{M_\odot\ year^{-1}}$ are commonly encountered (Section 3.2.2). For a low-mass pre-MS (main sequence) Pop III star with $M_* = 1\,\mathrm{M_\odot}$ and $R_* = 20\,\mathrm{R_\odot}$ (Omukai & Palla 2003), this corresponds to accretion luminosities in the range of $L_{\rm acc} \sim 2{,}000$–$20{,}000\,\mathrm{L_\odot}$, which is much higher than the MS luminosities of these same stars. Accretion onto more massive Pop III protostars produces even higher luminosities, with $L_{\rm acc}$ scaling approximately as $L_{\rm acc} \propto M_*^{0.73}$ for fixed $\dot{M}$ (Smith et al. 2012).

Despite this, the effect of this radiation on the surrounding gas is relatively modest. Rapidly accreting Pop III protostars have photospheric temperatures of at most ~6,000 K (Stahler et al. 1986b, Omukai & Palla 2003) and, hence, radiate most of their energy at visible and infrared wavelengths where the continuum opacity of metal-free gas is very small (Mayer & Duschl 2005). Therefore, only a tiny fraction of the radiated energy is absorbed by the gas in the minihalo, with most escaping into the intergalactic medium (IGM). The heating that this provides to the minihalo gas moderately reduces its propensity to fragment (Smith et al. 2011) but otherwise has little impact on its evolution.

4.1.2. Stellar luminosity. At later times, once the newly formed Pop III stars have joined the MS, their intrinsic luminosity becomes more important than their accretion luminosity, specifically when talking about high-mass stars above $\sim 10\,\mathrm{M_\odot}$. Even on the MS, if the stars continue to accrete, the geometry of the flow has strong impact on their internal structure (Hosokawa et al. 2012b). There are two extreme cases that we can consider. Accretion through a geometrically thin disk-like structure delivers fresh material with relatively low entropy. It settles onto the stellar surface with the same specific entropy as the photosphere, which allows the Pop III star to retain a relatively compact configuration. The total stellar luminosity L_* relates to the radius R_* and effective surface temperature T_* as

$$L_* = 4\pi\sigma_{\rm SB} R_*^2 T_*^4\,, \qquad 19.$$

with $\sigma_{\rm SB}$ being the Stefan–Boltzmann constant. Consequently, a star subjected to cold disk accretion is relatively hot and compact. If the accretion flow is spherically symmetric, the infalling material retains the entropy produced in the accretion shock front, and the star responds to the delivery of high-entropy material by increasing its radius. This implies that the surface temperature must drop, and so stars experiencing spherical hot accretion tend to be relatively cool and bloated (Hosokawa et al. 2012b).

The exact values of the stellar radius R_*, effective surface temperature T_*, and total luminosity $L = L_{\rm acc} + L_*$ vary with time and strongly depend on the instantaneous accretion rate $\dot{M}$ and mass M_* (Hosokawa & Omukai 2009; Hosokawa et al. 2010, 2012a; Haemmerlé et al. 2018). However, once the stars have finished accreting, they settle onto a well-defined MS. Pop III stars on this MS are more compact than their present-day counterparts, with a mass–radius relation at the high-mass end given roughly by

$$R_* \sim 0.3\ \mathrm{R_\odot} \left(\frac{M_*}{\mathrm{M_\odot}}\right)^{0.6}. \qquad 20.$$

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They are therefore somewhat hotter than their present-day counterparts (Maeder & Meynet 2012). We illustrate key properties of Pop III stars for masses ranging from $M_* = 1.7\,\mathrm{M}_\odot$ to $500\,\mathrm{M}_\odot$ in **Figure 8**. The values are based on the stellar evolution models computed by Murphy et al. (2021a,b) and S. Martinet, G. Maynet, S. Ekström, C. Georgy, R. Hirschi, et al. (in preparation) using the Geneva code (Eggenberger et al. 2008). In **Figure 8*a***, we plot the location of the stars in the $\log_{10} L$–$\log_{10} T$ diagram during the MS and post-MS evolution. For comparison, we also provide the zero-age main sequence (ZAMS; see, e.g., Kippenhahn et al. 2012) of solar metallicity stars (from Ekström et al. 2012) in the mass range from $0.8\,\mathrm{M}_\odot$ to $120\,\mathrm{M}_\odot$. This demonstrates that primordial and present-day stars of the same initial mass have similar total luminosity during the MS evolution, before mass loss becomes important for solar metallicity stars. However, due to their much higher compactness, T_* is considerably larger. In **Figure 8*b***, we visualize the stellar lifetime, as well as the stellar radius (with the blue line justifying Equation 20), the total luminosity, and the temperature as functions of mass at various evolutionary phases. The largest symbol depicts the stars at the ZAMS, whereas the smallest symbol shows the stars in their giant phase. Altogether, we provide these properties in steps of 10% of the total stellar lifetime. We note that different stellar evolutionary models and codes (e.g., SEVN; see Spera et al. 2022) lead to very similar numbers. Further information can be found in the **Supplemental Text** together with **Supplemental Tables 1 and 2** showing the fluxes of ionizing and nonionizing photons, and the total number of photons produced per baryon, including also population-averaged values.

Zero-age main sequence (ZAMS): onset of core hydrogen burning; as the star evolves on the main sequence it becomes somewhat hotter and brighter

Supplemental Material >

In summary, the existing models strongly suggest that high-mass Pop III stars are very compact with high photospheric temperatures (Omukai & Palla 2003), and they therefore radiate a large fraction of their energy at ultraviolet wavelengths. Once Pop III stars with masses greater than about 10–$20\,\mathrm{M}_\odot$ have formed, they become powerful sources of ionizing photons. These photons can have a profound effect on the surrounding gas, as we explore below in Sections 4.2 and 4.3.

4.1.3. Influence of rapid accretion on the stellar spectrum. The simple picture sketched above assumes that once a Pop III star reaches the MS, it remains there in a compact state. This is a reasonable assumption if the accretion rate onto the star is less than $\sim 10^{-2}\,\mathrm{M}_\odot\,\mathrm{year}^{-1}$ (e.g., Hosokawa & Omukai 2009, Hosokawa et al. 2010, Maeder & Meynet 2012). However, if the accretion rate onto the star becomes larger, this triggers the expansion of the outer envelope of the star to a red supergiant state, in which the stellar radius is given by (Hosokawa et al. 2012a, 2013; Haemmerlé et al. 2018)

$$R_* = 2.6 \times 10^2\,\mathrm{R}_\odot \left(\frac{M_*}{\mathrm{M}_\odot}\right)^{1/2}. \qquad 21.$$

This can be analytically derived by taking the Eddington luminosity, which is realistic for photon-dominated massive stars (e.g., Kippenhahn et al. 2012), adopting an effective surface temperature of $T_* \sim 5{,}000$ K, which is appropriate for red supergiants and which turns out to be roughly constant across a wide range of masses, and inserting both values into Equation 19. For comparison, the radius of a ZAMS Pop III star with $M_* = 100\,\mathrm{M}_\odot$ is $R_* \approx 5\,\mathrm{R}_\odot$ according to Equation 20. The intrinsic luminosity of the star is not strongly affected by this change in its structure, and Equation 19 implies that the increase of R_* leads to a substantial drop of T_* down to values of $\sim$5,000 K, remaining relatively constant across a wide range of masses. This leads to a massive drop in the ionizing photon output of the star (Hosokawa et al. 2016). If the Pop III star is located in an environment where an accretion rate of $10^{-2}\,\mathrm{M}_\odot\,\mathrm{year}^{-1}$ or above onto the star can be maintained for an extended period, the star never develops an HII region. Therefore, photoionization feedback never has the opportunity to interfere with accretion onto the star, and it can grow to become extremely massive (as discussed in Section 3.4).

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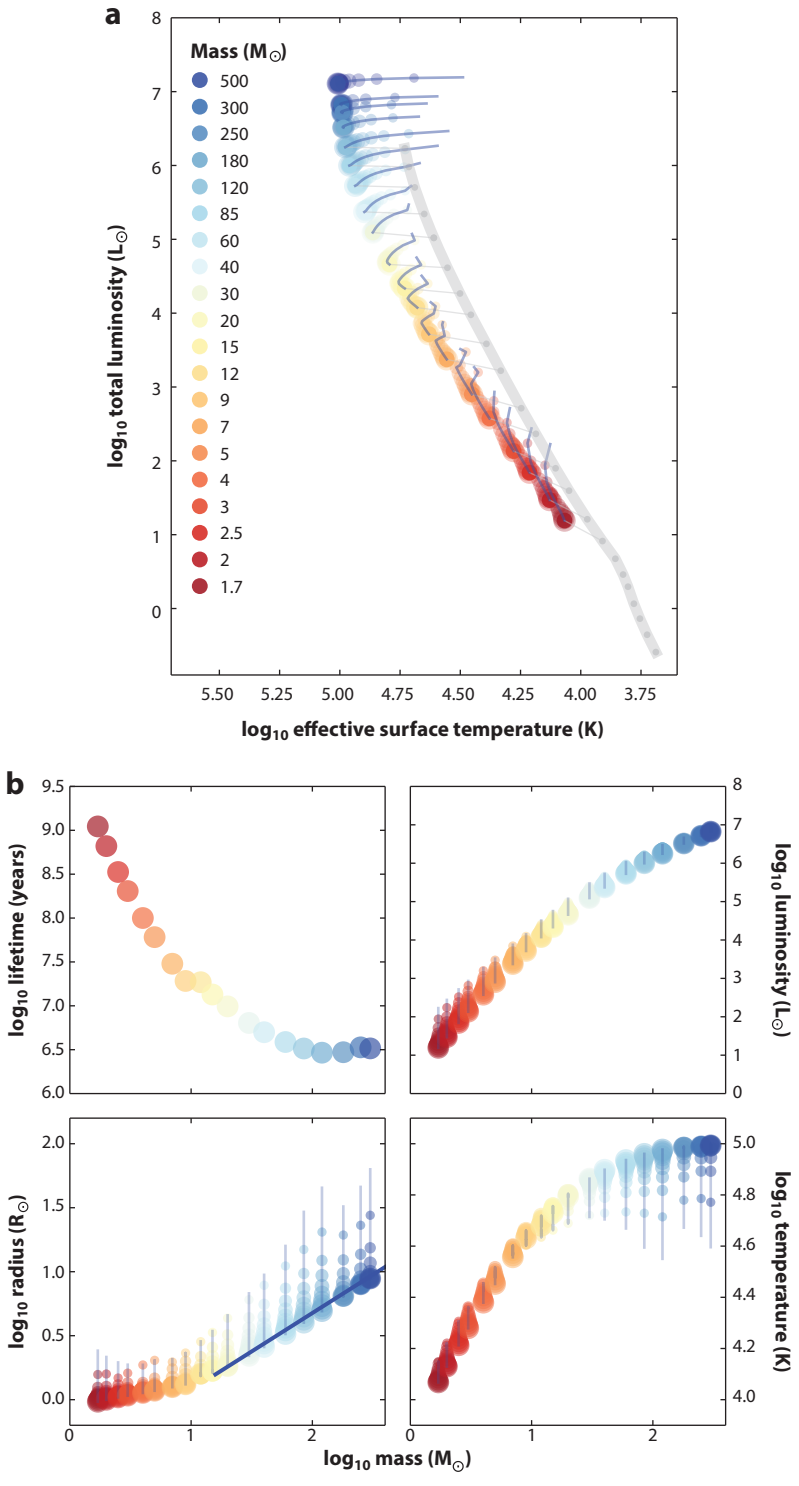

Figure 8

Summary of key properties of Pop III stars in the mass range of $M_* = 1.7$–$500\,M_\odot$. (*a*) The stellar loci in the Hertzsprung–Russell diagram ($\log_{10} L$ versus $\log_{10} T$) across their MS and post-MS phases. For comparison, we also show the ZAMS of solar metallicity stars in the range of 0.8–120 $M_\odot$ (in *gray*). (*b*) Numbers are presented for stellar lifetime, stellar radius, total luminosity, and effective surface temperature, each as a function of mass. The symbol size indicates the evolutionary stage, with the largest one indicating the ZAMS and the smallest the post-MS giant phase, and with a step size that corresponds to 10% of the total stellar lifetime. The solid blue line illustrates the mass–radius relation for massive MS Pop III stars (Equation 20). The underlying data are taken from Murphy et al. (2021a,b) and S. Martinet, G. Maynet, S. Ekström, C. Georgy, R. Hirschi, et al. (in preparation) for zero metallicity, and from Ekström et al. (2012) for solar metallicity. More details are provided in the **Supplemental Text**. Abbreviations: MS, main sequence; Pop III, Population III; ZAMS, zero-age main sequence.

Supplemental Material >

Furthermore, even if the mean accretion rate onto the star is less than 10^{-2} $M_\odot$ year^{-1}, it can still grow to reach a supergiant state if the accretion is highly episodic, with periods in which the accretion rate exceeds the critical value (Vorobyov et al. 2013). In this case, a period of rapid accretion triggers the expansion of the outer envelope of the star, but once the accretion rate falls below the critical value, the star contracts back to its original size. This process can happen repeatedly during the growth of the star (Hosokawa et al. 2016) and leads to the repeated appearance and disappearance of the surrounding HII region, in a fashion similar to the flickering found in simulations (e.g., Peters et al. 2010, Galván-Madrid et al. 2011) and observations (e.g., De Pree et al. 2015) of present-day HII regions.

How common this behavior is in real systems depends on how often young Pop III stars encounter such high accretion rates. This is difficult to predict robustly, because it depends not only on the larger-scale environment in the halo but also on the details of the disk fragmentation, the proportion of fragments that merge, etc. Simulations of this yield numerically converged results only when run with very high resolution (Prole et al. 2022a), and as yet very few such simulations have been run for long enough to also capture the formation of massive stars (see the discussion in Section 3.2.3).

4.2. Small-Scale Impact

In this section, we summarize the impact that feedback from Pop III stars has on gas close to the stars, which here we take to mean gas within the same dark matter halo. The impact of Pop III stellar feedback on much larger scales is discussed in Section 4.3 below.

4.2.1. Photoionization. As argued above, high-mass Pop III stars are thought to produce copious amounts of ionizing radiation. Consequently, we begin our discussion of stellar feedback by investigating the impact of photons with energies above 13.6 eV.

4.2.1.1. Impact on surroundings. The ionizing photons produced by a massive Pop III star are readily absorbed in the surrounding gas, leading to the formation of an HII region. The evolution of this HII region once it breaks out of the dense protostellar accretion disk has been modeled by a number of different groups (e.g., Whalen et al. 2004, Wise & Abel 2008, Hosokawa et al. 2016, Stacy et al. 2016) and is reasonably well understood. Because of the high densities found in the disk, the gas there resists photoionization, and so the HII region initially expands in a bipolar fashion above and below the disk midplane, as illustrated in **Figure 9**. The gas within the HII region is hot—temperatures of 1–2 $\times$ 10^4 K are typical (Whalen et al. 2004)—and highly overpressured compared to the surrounding neutral gas. It therefore expands outward, driving a shock in front of it and leaving behind a low-density cavity. The expansion of the HII region reduces the flow of gas onto the protostellar accretion disk, because gas can now reach the disk only by moving inward close to the midplane. In addition, the disk itself begins to be photoevaporated by the radiation from the star (McKee & Tan 2008). Together, these effects lead to the loss of the disk and the termination of accretion onto the star after a few 10^4 years (Hosokawa et al. 2016, Sugimura et al. 2020).

The final fate of the HII region depends upon its location. In a minihalo, the fact that the sound speed of the photoionized gas exceeds the escape velocity of the minihalo means that the HII region continues to expand for as long as the Pop III star continues to emit ionizing photons. For minihalos with masses close to $M_{\rm crit}$, the end result is that a large fraction of the gas mass is driven out beyond the virial radius (Kitayama et al. 2004, Whalen et al. 2004, Alvarez et al. 2006, Abel et al. 2007). In more massive minihalos, although the ionized gas is too warm to remain gravitationally bound to the minihalo, the larger size of the system means that a single HII region

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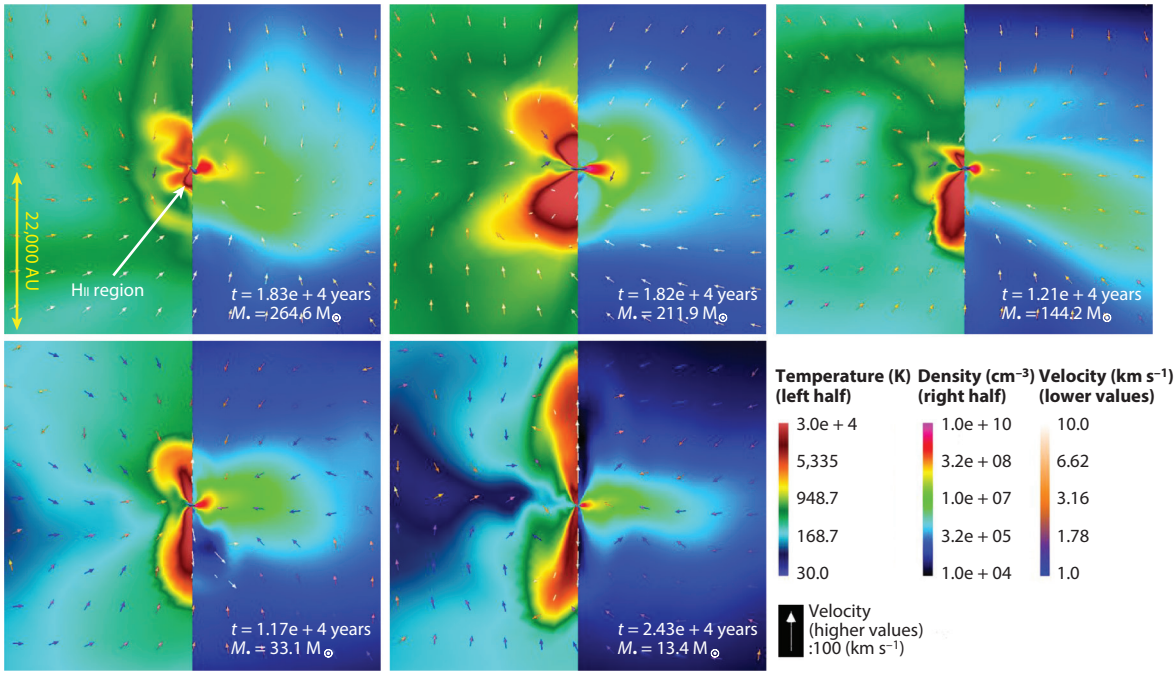

Figure 9

Illustration of the bipolar expansion of HII regions around massive Pop III stars. Five examples from the simulations of Hosokawa et al. (2016) are shown. In each case, the Pop III star is located at the center of the double panel and the temperature, density and velocity are shown in the plane containing the polar axis of the disk. Velocities below 10 km s^{-1} are denoted by color-coded arrows of fixed size; those above 10 km s^{-1} are denoted by white arrows, with lengths varying in proportion to the velocity. All five HII regions are shown at the time at which their polar extension first reaches 10^4 AU. This time is indicated in the figure for each HII region, along with the mass of the Pop III star at this moment. Figure adapted from Hosokawa et al. (2016) with permission from ApJ. Abbreviation: Pop III, Population III.

is unlikely to drive out all gas. Finally, in protogalaxies with $v_{esc} \sim 20$ km s^{-1} or more, the ionized gas remains gravitationally bound to the protogalaxy, which therefore retains most of the gas. For a similar reason, the time-averaged escape fraction of ionizing photons from the HII region, i.e., the fraction of photons that can travel into the IGM, also depends strongly on the location of the HII region. In small minihalos, it can reach values of 50% or more (see, e.g., Kitayama et al. 2004, Alvarez et al. 2006, Wise et al. 2014, Schauer et al. 2015), whereas in more massive galaxies, a broader range of values is possible (see, e.g., Paardekooper et al. 2013, Dayal & Ferrara 2018).

4.2.1.2. HII region trapping. An additional complication has been highlighted by Jaura et al. (2022). Most numerical studies of the growth of Pop III HII regions implicitly assume that the HII region has already escaped from the dense protostellar accretion disk by choosing an injection region for the ionizing photons that is larger than the disk scale height near the star.[3] Jaura et al. (2022) show that if one instead injects the photons at the location of the star itself, then the HII

[3]The studies by Stacy et al. (2016) and Sugimura et al. (2020) do not make this assumption. Instead, they each adopt a different subgrid model to treat the behavior of the HII region on scales that are not resolved in their calculation. However, neither of these subgrid models properly captures the behavior of the HII region near the star, as discussed by Jaura et al. (2022).

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region remains trapped in the dense gas and does not break out of the disk. The physical reason for this is that at the high densities found close to the star, the Strömgren radius is much smaller than the Bondi–Parker radius at which the thermal pressure of the ionized gas first starts to dominate over the gravitational attraction of the star. Therefore, the ionized gas remains gravitationally bound to the star, preventing it from flowing outward and escaping (Keto 2007). This result holds regardless of whether or not one accounts for the radiation pressure exerted on the gas by the ionizing photons, as this is too weak to overcome gravity close to the star. However, Jaura et al. (2022) also note that there are two important physical processes that are not included in their model that might change this picture. First, their simulations do not include a magnetic field and, hence, are unable to capture the formation of a magnetically powered jet or outflow. If such an outflow forms, then this removes dense gas from the vicinity of the star, potentially allowing the HII region to break out from the disk (McKee & Tan 2008). Second, the high optical depth of the accretion disk to Lyman-α radiation means that any of the photons produced by the star or in the HII region scatter repeatedly before escaping. The resulting radiation pressure is orders of magnitude greater than the direct radiation pressure of the ionizing photons and, hence, may be sufficient to overcome the gravitational attraction of the star (McKee & Tan 2008, Jaura et al. 2022). Modeling this requires a careful treatment of the interaction between the radiation field and the gas in the dense disk environment and should be a priority for future simulations.

4.2.2. Photodissociation. As well as producing a large number of ionizing photons, massive Pop III stars also emit many photons in the LW bands of H_2 (11.2–13.6 eV). Because of the crucial importance of H_2 cooling in Pop III star formation, photodissociation of H_2 by these photons is potentially an important form of negative feedback.

At high densities (e.g., in the immediate vicinity of the disk), the photodissociation region (PDR) created by these photons is restricted to a narrow region just outside of the HII region. This is because as dissociating photons penetrate into the gas ahead of the ionization front (I-front), they build up a dense layer of atomic hydrogen. Once the column density of this layer exceeds $N_{\rm H} \sim 10^{24}\,{\rm cm}^{-2}$, absorption of LW photons in the damping wings of the atomic hydrogen Lyman series lines starts to be highly effective (Wolcott-Green & Haiman 2011). This shields the gas ahead of the dense atomic layer and limits its maximum column density to $\sim 10^{25}\,{\rm cm}^{-2}$. In the density regime where three-body H_2 formation is important ($n > 10^{11}\,{\rm cm}^{-3}$; see **Figure 3**), the maximum spatial extent of this layer is only $L_{\rm max} \sim 10^{14}\,n_{11}^{-1}\,{\rm cm} \approx 10\,n_{11}^{-1}\,{\rm AU}$, where n_{11} is the gas density in units of $10^{11}\,{\rm cm}^{-3}$. Photodissociation of H_2 in this narrow PDR leads to a dramatic increase in its temperature, driven primarily by H_2 formation heating (Susa 2013). Unfortunately, the impact of this temperature increase remains unclear. Some authors (Susa 2013, Susa et al. 2014, Stacy et al. 2016) report that it drives a pressure wave outward that reduces the gas density near the star and dramatically reduces its accretion rate even in the absence of photoionization feedback. By contrast, Hosokawa et al. (2016) find it to have a much more limited effect and argue that photoionization is the dominant process. However, none of these studies accounts for the shielding of H_2 by atomic hydrogen, and hence they all overestimate the amount of dense gas affected by photodissociation feedback.

At lower densities, shielding of H_2 by atomic hydrogen becomes ineffective, as the required shell thickness becomes implausibly large. For example, at a density of $n_{\rm H} = 10^8\,{\rm cm}^{-3}$, producing a column density of $10^{25}\,{\rm cm}^{-2}$ requires a shell thickness of ~7,000 AU, which is several times larger than the entire physical extent of the region with mean density $>10^8\,{\rm cm}^{-3}$ (see **Figure 4**). Therefore, the PDR is no longer restricted to the gas close to the HII region and instead can expand to large distances within $\sim 10^4$ years. Gas in the PDR is not strongly heated by photodissociation, which is therefore ineffective at driving gas out of the minihalo. However, the loss of the

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main coolant from the gas leads to it heating up to a temperature of 5,000–10,000 K as it collapses (Susa et al. 2014). Over time, this heating reduces the flow of gas to the center of the minihalo, limiting the amount of mass available for accretion even in the absence of effective photoionization feedback.

4.2.3. Outflows. Protostellar outflows, in the form of both highly collimated jets and also wide-angle winds, are a ubiquitous feature of present-day star-forming regions (Bally 2016). These outflows are magnetocentrifugal in nature. They are powered by the rotation of the protostellar accretion disk and channeled and collimated by the toroidal magnetic field built up by the rotation of the disk (Pudritz & Ray 2019). Because we now understand that a dynamically significant magnetic field can be present even in the earliest minihalos (Schober et al. 2012a; see also Section 3.3.2), an obvious question is whether the formation of Pop III stars is also associated with the presence of magnetized outflows.

In practice, the answer to this question appears to depend on what one assumes regarding the initial geometry of the field. Machida et al. (2008a), Machida & Doi (2013), and Sadanari et al. (2021) simulate the collapse of primordial gas clouds threaded by an initially uniform magnetic field and show that protostellar outflows are a common outcome unless the initial magnetic field strength is very small. In contrast, the simulations performed by Sharda et al. (2020) and Prole et al. (2022b) that start with a tangled magnetic field of the kind that we would expect to result from the turbulent dynamo find no evidence for the launching of jets or outflows. Whether this means that Pop III protostars do not produce outflows at all or simply that outflows are launched at a later point in the evolution of the system than is followed by these simulations remains to be determined.

The other class of stellar outflows that are potentially of interest are winds driven by the Pop III stars themselves. At solar metallicity, massive stars can produce highly energetic winds that can have a profound impact on the surrounding gas (see, e.g., Rahner et al. 2017). However, these winds are driven by the absorption of stellar photons by the lines of highly ionized metals and we expect the wind strength to be a function of metallicity (Kudritzki 2002). They become weak or nonexistent for zero metallicity (Krtička & Kubát 2006, 2009). Accounting for stellar rotation does not significantly change this picture: Stars rotating very close to their break-up velocity can drive strong winds but lose so much angular momentum in the process that they do so only very briefly (Ekström et al. 2008). Stellar wind feedback therefore does not appear to be important for understanding the formation of Pop III stars.

4.3. Large-Scale Impact

In addition to the profound impact that feedback from the first stars has on the gas in their parent minihalos, feedback from these stars also strongly affects gas on much larger scales, in both the IGM and neighboring protogalaxies. In this section, we review the most important of these impacts.

4.3.1. Photoionization. Ultraviolet photons that manage to escape from the halos hosting Pop III star formation start to ionize the surrounding IGM. Prior to cosmic reionization, the Universe is highly opaque to these types of photons, and so this is essentially a local process: Individual star-forming protogalaxies produce distinct ionized regions in the IGM, but no large-scale ionizing background builds up until after these individual regions begin to overlap, an event that does not occur until long after Pop II stars have taken over as the dominant source of ionizing photons (Hartwig et al. 2022).

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The impact of photoionization feedback from Pop III star formation on nearby minihalos and atomic cooling halos depends on their evolutionary state and on the duration of the feedback. Systems in which gas has already cooled and is in the process of forming stars are difficult to disrupt with external photoionization feedback, because the time taken for an I-front to propagate through the system is much longer than the collapse time of the gas once the gas density exceeds $n \sim 100\,\mathrm{cm}^{-3}$ (see, e.g., Whalen et al. 2008). However, systems that have not yet gathered together much cool gas can be strongly affected. Visbal et al. (2017) show for a constant ionizing flux that photoionization increases the minimum virial temperature required for collapse to $T_{\mathrm{vir}} \sim 40{,}000$ K if the ionizing flux exceeds $F_{\mathrm{crit}} \sim 10^6$ photons $\mathrm{s}^{-1}\,\mathrm{cm}^{-2}$. For smaller values, photoionization is of limited importance, although cooling is suppressed in halos with $T_{\mathrm{vir}} < 10^4$ K by the LW radiation that Visbal et al. (2017) assume accompanies the ionizing radiation (see Section 4.3.2 below). For Pop III minihalos with star-formation efficiencies typical of those found in simulations, the ionizing flux exceeds F_{crit} only at distances of a few kiloparsecs or less, even when assuming an escape fraction of 100%. Furthermore, the duration of Pop III star formation in individual minihalos is short, owing to the strong effects of photoionization and SN feedback on small scales. Therefore, cooling is only suppressed for a brief period of a few megayears. Over longer timescales, the effect of ionizing feedback within this region can actually be positive, because the residual fractional ionization in the recombining region promotes H_2 formation (O'Shea et al. 2005). Overall, we do not expect photoionization by Pop III-dominated sources to be the main source of feedback at high redshifts. Feedback from Pop II-dominated protogalaxies on surrounding minihalos is potentially more important, but a thorough discussion of this topic is outside of the scope of this review.

4.3.2. Photodissociation. As discussed in Section 4.2.2, massive Pop III stars produce large numbers of LW band photons that can photodissociate H_2. A large fraction of these escape from their parental minihalos (Kitayama et al. 2004; Schauer et al. 2015, 2017a), and they can propagate to large distances through the IGM owing to its very low opacity below 13.6 eV (Haiman et al. 2000). The onset of Pop III star formation is therefore quickly followed by the buildup of an extragalactic LW background that pervades the Universe. This photodissociates H_2 in newly assembled minihalos, making it harder for the gas to cool (Haiman et al. 1997). Once the strength of the background exceeds a critical value, H_2 cooling is almost entirely suppressed, preventing gas in minihalos from collapsing and forming Pop III stars.

The strength of the LW background required in order to suppress H_2 cooling in a minihalo can be estimated by a simple timescale argument (Oh & Haiman 2002). Models for the buildup of H_2 in primordial gas demonstrate that most molecules form within the first few recombination times, with the H_2 formation rate dropping substantially at later times owing to the loss of the required electrons and protons from the gas (Tegmark et al. 1997, Oh & Haiman 2002, Glover 2013). The LW background therefore only significantly suppresses H_2 formation once the photodissociation timescale, $t_{\mathrm{dis}} = 1/k_{\mathrm{dis}}$, becomes comparable with the recombination timescale, $t_{\mathrm{rec}} = 1/(k_{\mathrm{rec}} n x_0)$, where k_{dis} and k_{rec} are the photodissociation and radiative recombination rate coefficients, respectively; n is the characteristic number density of the gas; and x_0 is its initial fractional ionization. For the conditions appropriate for a minihalo close to the minimum mass scale for collapse (see Section 3.1), that is $T \sim 2{,}000$ K and $x_0 \sim 2 \times 10^{-4}$, this yields a critical value (Oh & Haiman 2002) of

$$\left(\frac{J_{21}}{n}\right)_{\mathrm{crit}} \approx 10^{-4} f_{\mathrm{sh}}, \qquad 22.$$

above which we expect the LW background to significantly affect the minihalo. Here, J_{21} is the strength of the background at the Lyman limit in units of $10^{-21}\,\mathrm{erg\,s^{-1}\,cm^{-2}\,Hz^{-1}\,sr^{-1}}$, n is the density of the gas at the center of the minihalo, and f_{sh} is the self-shielding factor, i.e., the ratio of

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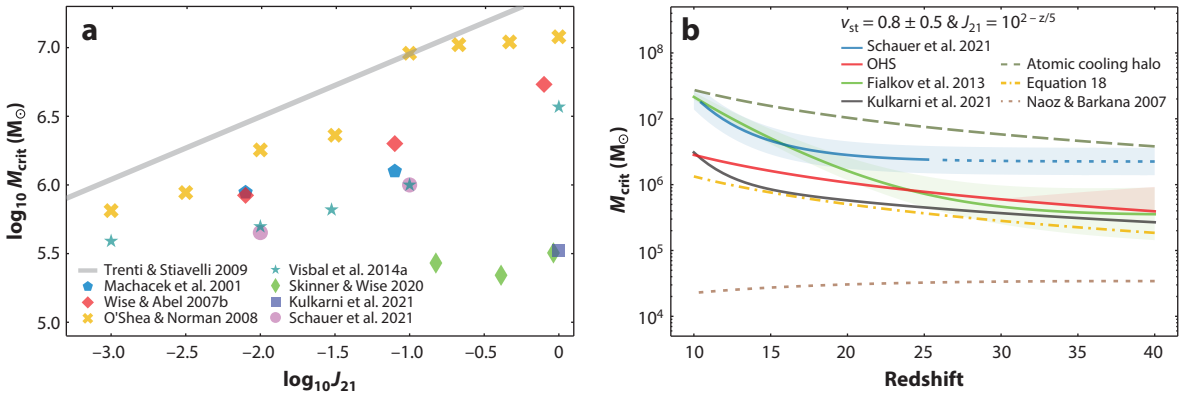

Figure 10

(*a*) Minimum minihalo mass required for Pop III star formation (M_{crit}) plotted as a function of J_{21}, the strength of the LW background in units of 10^{-21} erg s^{-1} cm^{-2} Hz^{-1} sr^{-1}. The values shown are taken from studies by Trenti & Stiavelli (2009, *gray line*), Machacek et al. (2001, *blue pentagons*), Wise & Abel (2007b, *red diamonds*), O'Shea & Norman (2008, *yellow crosses*), Visbal et al. (2014a, *cyan stars*), Skinner & Wise (2020, *green diamonds*), Kulkarni et al. (2021, *navy squares*), and Schauer et al. (2021, *lavender dots*). For Trenti & Stiavelli (2009), we show the values computed assuming $z = 20$. Skinner & Wise (2020) find H_2 cooling and star formation occurring in minihalos with a broad range of masses for $10^{-1} < J_{21} < 1$; we show only the few least massive points from their study here. Finally, for Schauer et al. (2021), we show the values they report for the lowest mass at which minihalos form stars. The average mass at which minihalos first form stars (their M_{ave}) is about a factor of three larger than this minimum value. (*b*) Evolution of M_{crit} with redshift for a selection of models that account for the combined effects of baryonic streaming and the LW background. The lines are computed for a streaming velocity $v_{st} = 0.8$ (in units of the rms value), and the shading indicates the impact of varying this by ± 0.5. For the time evolution of the LW background, we adopt a simple fit to the values computed by Greif & Bromm (2006). The meaning of the various lines is the same as in **Figure 1**. Abbreviations: LW, Lyman–Werner; OHS, O'Shea–Hummel–Stacy; Pop III, Population III.

the H_2 photodissociation rate to its value in the absence of self-shielding. Numerical simulations (e.g., Machacek et al. 2001, Yoshida et al. 2003, Kulkarni et al. 2021) indicate that $n \sim 10\,\mathrm{cm}^{-3}$ is an appropriate value for the central density in the earliest minihalos, and so in the absence of self-shielding, we expect the LW background to become important once its strength exceeds a critical value of around $J_{21,crit} \sim 10^{-3}$. In this case, the LW background reduces the peak H_2 abundance by a factor proportional to $t_{dis}/t_{rec} \propto J_{21}^{-1}$, thereby increasing the minimum virial temperature T_{min} required for efficient H_2 cooling. At typical minihalo temperatures, the H_2 cooling rate scales approximately as $\Lambda_{H_2} \propto T^4$, and the cooling time scales as $t_{cool} \propto (T^3 x_{H_2})^{-1}$. Therefore, for the cooling time to remain approximately constant as J_{21} increases, we must have $T_{min} \propto J_{21}^{1/3}$. Furthermore, because $M_{crit} \propto T_{min}^{3/2}$, this implies that the minimum mass scale for collapse must increase as $M_{crit} \propto J_{21}^{1/2}$ for $J_{21} > J_{21,crit}$.

In **Figure 10*a***, we show a compilation of predictions for how M_{crit} varies as a function of J_{21} taken from a number of studies in the literature. These studies range from simple semianalytical models (Trenti & Stiavelli 2009, Visbal et al. 2014a) to detailed numerical simulations both with (Skinner & Wise 2020, Kulkarni et al. 2021, Schauer et al. 2021) and without (Machacek et al. 2001, Wise & Abel 2007b, O'Shea & Norman 2008) H_2 self-shielding. Although there is considerable scatter,[4] we see that the semianalytical models and the simulations without H_2 self-shielding yield results that broadly agree with our expectations based on the discussion above. However, the

[4]This scatter likely results from a combination of differences in the microphysical treatment (e.g., selection of reactions, choice of reaction rate coefficients and cooling function) and differences in the definition of M_{crit} in the different studies.

situation becomes more complicated once we account for H_2 self-shielding. Schauer et al. (2021) include this in their simulations but do not find it to have a large effect. They report values of $M_{\rm crit}$ that are a factor of 2–3 smaller than those predicted by simulations that do not account for self-shielding, but whether this difference is due to the inclusion of self-shielding or due to other differences in the chemical or thermal treatment is unclear. In contrast, Skinner & Wise (2020) and Kulkarni et al. (2021) find H_2 self-shielding to have a profound impact on $M_{\rm crit}$, keeping it below $10^7\,M_\odot$ for LW background strengths of up to $J_{21} \sim 10$. If true, these results would imply that the LW background does not play an important role in regulating high redshift star formation, because the mean strength of the background does not exceed $J_{21} = 1$ until after Pop II star formation has already started to dominate (see, e.g., Trenti & Stiavelli 2009, Ahn et al. 2009).

Another view of the current uncertainty in the impact of LW feedback is given in **Figure 10*b***, where we show the redshift evolution of $M_{\rm crit}$ in several different models, assuming a streaming velocity $v_{\rm st} = 0.8$ (in units of the root-mean-squared value) and an LW background that evolves as $J_{21} = 10^{2-z/5}$, which is a reasonable fit at $z > 10$ to the values computed by Greif & Bromm (2006). At $z > 25$, the LW background is very weak, and the evolution of $M_{\rm crit}$ in all of the models is essentially the same as in the $J_{21} = 0$ case discussed earlier (see Section 3.1.2 and **Figure 1**). At $z < 25$, however, the growing background drives a substantial increase in $M_{\rm crit}$ in the Schauer et al. (2021) and F13 models but has little impact on it in the Kulkarni et al. (2021) and OHS models.

Figure 10 also demonstrates that in halos more massive than $\sim 10^7\,M_\odot$ at $z = 20$, $M_{\rm crit}$ becomes highly insensitive to J_{21} even in the absence of effective H_2 self-shielding. The reason for this change of behavior is that these halos have high enough virial temperatures to allow them to cool initially via Lyman-α cooling. In these atomic cooling halos, the gas can reach much higher densities without the need for H_2 cooling, meaning that a much stronger LW background is needed to completely suppress H_2 formation. The key density in this case is $n_{\rm crit} \sim 10^4\,{\rm cm}^{-3}$, the critical density above which the rotational and vibrational level populations of H_2 start to approach their local thermodynamical equilibrium (LTE) values. Above $n_{\rm crit}$, the H_2 cooling time becomes independent of density, whereas the free-fall collapse time continues to decrease with increasing density. In addition, collisional dissociation of H_2 also becomes more effective in gas with $n > n_{\rm crit}$. Together, these facts lead to a bifurcation in the behavior of atomic cooling halos depending on the amount of H_2 they have managed to form by the time their density reaches $n_{\rm crit}$. If they have formed enough H_2 to provide effective cooling, then the gas remains cold as it collapses, with its subsequent evolution being very much the same as in the standard Pop III formation scenario (Section 3.2). However, if H_2 cooling has not become effective by the time the gas reaches $n_{\rm crit}$, it will never become so at higher densities. Instead, the gas remains warm, with cooling arising primarily from bound–bound transitions in atomic hydrogen and from H^- formation (Omukai 2001, Schleicher et al. 2010).

The combined effect of the higher temperatures and higher characteristic densities encountered in atomic cooling halos is a dramatic increase in the value of $J_{21,\rm crit}$. One-zone models of the thermal and chemical evolution of highly irradiated gas in atomic cooling halos typically find $J_{21,\rm crit} \sim 1{,}000$ if the spectrum of the LW background is dominated by Pop III stars (see, e.g., Sugimura et al. 2014; Glover 2015a,b). Three-dimensional models find even larger values, $J_{21,\rm crit} \sim 10^4$ (Shang et al. 2010, Hartwig et al. 2015, Latif et al. 2015), although the scatter between different simulations can be considerable.

To put these numbers into context, we expect the mean strength of the LW background at reionization to be $J_{21} \sim 1$ (Haiman et al. 2000, Ahn et al. 2009). Therefore, the LW background is unable to suppress H_2 cooling in the vast majority of atomic cooling halos, although it does produce systematically warmer temperatures and higher accretion rates than would be the case without the background (Latif & Volonteri 2015, Regan & Downes 2018). Complete suppression

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of H_2 cooling occurs only in a small number of atomic cooling halos that happen to lie very close to a strong source of LW photons (Dijkstra et al. 2008, 2014; Agarwal et al. 2012, 2019; Visbal et al. 2014b; Regan et al. 2017) or that are undergoing strong dynamical heating from a succession of mergers (Regan et al. 2020). However, despite their rarity, these systems are of considerable interest as the possible formation sites of supermassive Pop III stars (see Section 3.4).

4.3.3. X-rays. Although Pop III stars themselves do not produce significant emission at X-ray wavelengths, we nevertheless expect Pop III star formation to be closely associated with the production of X-rays. We discuss the origin of this emission as well as its impact on the surrounding gas.

4.3.3.1. Stellar sources of high-redshift X-rays. Three main mechanisms potentially contribute to the X-ray emission from Pop III stars. First, if some fraction of these stars are located in tight binaries, then these systems can become high-mass X-ray binaries (HMXBs, see, e.g., Mirabel et al. 2011). Second, hot shocked gas in Pop III supernova remnants (SNRs) emits strongly at X-ray energies until it cools. Finally, Pop III SNRs may also produce X-rays indirectly, by acting as a source of cosmic ray electrons. Because of the high CMB energy density at high redshift, cosmic ray electrons with energies of up to a few hundred gigaelectronvolts lose most of their energy by inverse Compton scattering CMB photons, producing a power-law inverse Compton spectrum that extends up to X-ray energies (Oh 2001).

A useful metric for comparing the relative importance of these mechanisms is the X-ray luminosity they produce per unit star-formation rate, L_X/SFR. In the local Universe, at metallicities close to solar, HMXBs produce a luminosity in the 0.5–2-keV energy band given by $L_X/\mathrm{SFR} \approx 6 \times 10^{39}\,\mathrm{erg\,s^{-1}}\,(\mathrm{M_\odot\,year^{-1}})^{-1}$ (see, e.g., Glover & Brand 2003, Kaur et al. 2022), although there is roughly a factor of two scatter between different observational determinations. However, the contribution from SNRs[5] is less well determined. Estimates of the ratio of the X-ray luminosity to the mechanical luminosity range from values as small as 2×10^{-4} (e.g., Helfand & Moran 2001) to ones as large as a few percent (e.g., Smith et al. 2005, Franeck et al. 2022), albeit with substantial time variability and scatter from region to region. If we take a value near the midpoint of this range, $L_X/L_{\mathrm{mech}} = 10^{-3}$ as a representative average (Rogers & Pittard 2014, Franeck et al. 2022), then we find that $L_X/\mathrm{SFR} \approx 6 \times 10^{38}\,\mathrm{erg\,s^{-1}}\,(\mathrm{M_\odot\,year^{-1}})^{-1}$ for SNRs (Leitherer et al. 1999), i.e., around 10% of the HMXB contribution. Therefore, on average in the local Universe, X-rays from HMXBs dominate over those produced by hot gas in SNRs. Accounting for the inverse Compton contribution does not substantially change this picture (Oh 2001).

How do these conclusions change as we move to higher redshift? Both models and observations show that HMXBs produce X-rays more efficiently as the metallicity drops, with L_X/SFR increasing by an order of magnitude or more as the metallicity decreases from $1\,\mathrm{Z_\odot}$ to $0.03\,\mathrm{Z_\odot}$ (Kaur et al. 2022). At even lower metallicities, we have no direct observational constraints, and theoretical models disagree on whether L_X/SFR asymptotes to a constant value (Fragos et al. 2013) or continues to increase (Brorby et al. 2016). It is also unclear whether this finding of a higher L_X/SFR remains valid if we extrapolate all the way down to $Z = 0$, with results from one study indicating that HMXBs formed from Pop III stars may actually be less effective at producing X-rays than in

[5]We do not consider stellar winds here, as even at solar metallicity, these do not succeed in producing large X-ray luminosities (Wang & Helfand 1991, Harper-Clark & Murray 2009), due to some combination of the escape of hot gas from wind-driven bubbles and efficient cooling of the gas at the bubble boundary (Lancaster et al. 2021). At low or zero metallicity, stellar winds are much weaker and, hence, are even less effective at producing X-rays.

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the solar metallicity case (Sartorio et al. 2023). Turning our attention to SNRs, we note that we do not expect the fraction of the energy they radiate as X-rays to be particularly sensitive to the metallicity. However, if the Pop III IMF is dominated by massive stars (see Section 3.2.3), then the number of SNe that explode per unit star-formation rate is a factor of a few larger than in the local Universe. In addition, if a significant number of Pop III stars explode as hypernovae (HNe) or pair-instability SNe (PISNe; see Section 4.3.4 below), then the energy available per explosion is also larger, implying a higher X-ray luminosity even if L_X/L_{mech} remains constant. In addition, the inverse Compton contribution is potentially far more important here than at low redshift. For a typical SN, around 0.1–0.2% of its initial kinetic energy is used to accelerate relativistic electrons (Lacki & Thompson 2013) and so if the bulk of this is converted to X-rays by inverse Compton scattering, the result is a contribution to L_X/SFR similar in size to that from hot gas.

In summary, if Pop III HMXBs are as efficient at producing X-rays as present-day HMXBs and if most Pop III SNe are standard core-collapse SNe (CCSNe) with explosion energies $E_{SN} \sim 10^{51}$ erg, then the majority of the X-ray emission associated with Pop III star formation comes from HMXBs. For this reason, it is this scenario that has attracted most attention in the literature, as we discuss in more detail below. However, scenarios in which Pop III SNe dominate the production of X-rays at high redshift cannot be completely ruled out (see, e.g., Ricotti 2016, for an example).

4.3.3.2. Impact of the high-redshift X-ray background. The X-rays produced by Pop III HMXBs and other sources can propagate to large distances through the IGM. The Universe is optically thin to X-ray photons with energies $E > 1.8[(1+z)/15]^{1/2}$ keV and the comoving photon mean free path exceeds 1 Mpc for photon energies $E > 0.25$ keV (Furlanetto et al. 2006). Therefore, once Pop III star formation begins, a pervasive X-ray background develops. It affects gas in high-redshift minihalos in two ways. First, it partially ionizes it, thereby catalyzing the formation of additional H_2 (Haiman et al. 2000). Second, it also heats the gas. The first of these effects acts as positive feedback, making cooling and star formation more likely, whereas the second one acts as negative feedback. Their relative importance depends on the density of the gas and the strength and spectral shape of the background. Heating dominates at low densities and can completely suppress the collapse of gas into low-mass minihalos if it is able to heat the IGM to a temperature above the virial temperature of the minihalo. The strength of the background required to accomplish this depends sensitively on the adopted low-energy cutoff, because the heating rate due to photons of energy E scales approximately as E^{-2} owing to the strong energy dependence of the photoionization cross section. For example, Hummel et al. (2015) take a minimum photon energy of 1 keV and estimate that a critical value of $J_{X,crit} \approx 10^{-21}$ erg s^{-1} cm^{-2} Hz^{-1} sr^{-1} at 1 keV is required to suppress the collapse of gas into a 10^6-$M_\odot$ halo. In contrast, Park et al. (2021a) adopt a minimum photon energy of 200 eV and find that a much weaker background is sufficient to suppress collapse, $J_{X,crit} \approx 10^{-23}$ erg s^{-1} cm^{-2} Hz^{-1} sr^{-1}.

At higher densities, the heating provided by the X-rays becomes unimportant in comparison to the increased cooling provided by the additional H_2. Consequently, the gas cools to lower temperatures than in the case without X-rays (Machacek et al. 2003, Hummel et al. 2015, Park et al. 2021a). The density at which the behavior changes lies in the range $n \sim 1$–100 cm^{-3}, with the value depending on the strength of the background. However, if the X-ray background is very strong, there is so much heating at low densities that the gas never reaches the positive feedback regime, and the feedback in this case is wholly negative.

At even higher densities, the gas becomes optically thick to the X-ray background, which therefore plays no direct role on scales comparable with the size of the protostellar accretion disk. However, its presence indirectly affects the behavior of the gas on these scales, as the lower temperatures that are reached on larger scales lead to a slower inflow of mass to the center of the

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halo. This may result in the formation of a less massive, more stable accretion disk (Hummel et al. 2015; Park et al. 2021a,b), which means fewer fragments, implying a more top-heavy IMF and lower stellar multiplicity.

Another interesting question is whether or not positive feedback from X-rays dominates over negative feedback from LW photons. The answer obviously depends on the relative strength of the two backgrounds (Park et al. 2021a), but models that self-consistently follow their buildup typically find that LW feedback dominates (see, e.g., Glover & Brand 2003).

Finally, the heating of the IGM produced by even a weak X-ray background has an important impact on its visibility in the 21-cm line of atomic hydrogen (Furlanetto et al. 2006). Prior to the onset of Pop III star formation, the IGM is colder than the CMB and, hence, the 21-cm signal is visible in absorption (see Section 5.1.3). However, once X-rays heat the IGM above the CMB temperature, the absorption signal vanishes, and the line instead becomes visible in emission. Measuring the redshift at which the transition from absorption to emission occurs therefore allows us to constrain the strength of the high-redshift X-ray background and may also allow us to distinguish between the contributions made to it by Pop III and Pop II sources (Mirocha et al. 2018).

4.3.4. Supernovae. In this section, we discuss how the final fate of a Pop III star depends on its initial mass, with a particular focus on identifying the range of masses that lead to SN explosions. We also briefly outline the impact that these explosions have on the surrounding gas.

4.3.4.1. The fate of massive Population III stars. How massive Pop III stars end their lives depends on the mass of their helium core at the moment that the star dies (Heger et al. 2003). For nonrotating stars with negligible mass loss, there is a simple mapping between helium core mass and initial mass, and so it is common to discuss the fates of these stars as a function of their initial mass. Below $M \sim 9\,\mathrm{M}_\odot$, Pop III stars end their lives as white dwarfs, supported by electron degeneracy pressure. Above this mass, electron degeneracy pressure is not sufficient to support the star. In the narrow mass range of $9\,\mathrm{M}_\odot < M < 10\,\mathrm{M}_\odot$, stars form degenerate oxygen–neon cores that collapse, whereas those with $10\,\mathrm{M}_\odot < M < 100\,\mathrm{M}_\odot$ form iron cores that also undergo collapse. In stars with $10\,\mathrm{M}_\odot < M < 25\,\mathrm{M}_\odot$, this core collapse triggers a SN explosion with an energy of around 10^{51} erg that leaves behind a neutron star remnant. For $25\,\mathrm{M}_\odot < M < 40\,\mathrm{M}_\odot$, a core-collapse explosion also occurs, but is unable to completely unbind the stellar envelope from the neutron star remnant. Instead, the inner portions of the envelope fall back onto the neutron star, causing it to collapse and form a black hole. The energy released into the surrounding environment depends on the extent of this fallback and can potentially be much smaller than 10^{51} erg, in which case we refer to the CCSN in question as a faint SN. For masses between $40\,\mathrm{M}_\odot$ and a somewhat uncertain upper limit in the range of 70–100 $\mathrm{M}_\odot$, no SN explosion occurs. Instead, the star simply collapses directly into a black hole at the end of its life.

For more massive stars, a new physical effect becomes important. These stars have such high central temperatures during the later stages of their evolution that they become susceptible to the so-called pair instability (Fowler & Hoyle 1964, El Eid et al. 1983, Ober et al. 1983, Bond et al. 1984, Heger & Woosley 2002). Briefly, as the temperature increases, so too does the fraction of photons in the core that have energies greater than 1.022 MeV, the rest mass of an electron–positron pair. Interaction of these photons with the surrounding nuclei can lead to their conversion into electron–positron pairs. Because the core is supported by radiation pressure, the loss of photons via this mechanism results in a reduction in the pressure and, hence, contraction of the core. This further increases the core temperature, making pair production more likely, and in sufficiently massive stars, this can become a runaway process triggering the dynamical implosion of the core (Woosley 2017). This implosion is terminated by the onset of rapid O and Si burning.

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Early calculations predicted an onset of the pair instability above a mass of around 100 $M_\odot$ (Heger et al. 2003), but more recent calculations find lower values of ~70 $M_\odot$ (Woosley 2017), albeit with some remaining uncertainty due to the uncertainty in the rate of the $^{12}C(\alpha, \gamma)^{16}O$ nuclear reaction (Farmer et al. 2019). In stars above this mass, but below 140 $M_\odot$, this leads to a pulsational instability in which the star undergoes a series of implosions, nuclear flashes, and reexpansions that disrupt the stellar envelope and part of the core. This process terminates once the core mass becomes too small for it to be susceptible to the pair instability, following which it evolves in the same fashion as a lower-mass star. The energy released by this instability ranges from $\ll 10^{51}$ erg to a few times 10^{51} erg, and the ejected mass also spans a large range of values (Woosley 2017).

Gamma-ray burst (GRB): a highly energetic burst of electromagnetic radiation that can last from tens of milliseconds to several hours

Above 140 $M_\odot$, pulsations do not occur, as the energy released in the initial nuclear flash completely disrupts the star, leaving no remnant behind. The resulting explosion is termed a PISN. It has a typical energy of around 10^{53} erg, which is two orders of magnitude larger than the value of a typical CCSN. Because all of the mass of the star is expelled in the ejecta, PISNe typically have very large metal yields. Finally, for Pop III stars more massive than 260 $M_\odot$, the pair instability occurs, but the energy released by the resulting rapid nuclear burning is not sufficient to disrupt the star, which instead simply collapses to a black hole (Heger & Woosley 2002).

Accounting for the effects of rotation has two main effects on the picture sketched above. First, rotation-driven mixing leads to rapidly rotating Pop III stars having larger helium core masses than nonrotating stars of the same mass (Yoon et al. 2012). Because the onset of the pair instability depends on the core mass, an important consequence of this is a reduction in the minimum mass required for a PISN, which can reach values as low as $M \sim 100\,M_\odot$ for rapidly rotating stars. Second, rapidly rotating Pop III stars can potentially explode as jet-driven SNe (see, e.g., Grimmett et al. 2021, and references therein). These are widely believed to be responsible for the unusually bright class of SNe known as HNe (see Umeda & Nomoto 2002), which have typical energies of 10^{52} erg. Rapid rotation is also a necessary (but not sufficient) condition for a SN to produce a gamma-ray burst (GRB; see Toma et al. 2016).

Unfortunately, the rotational properties of Pop III stars are poorly constrained. In order to measure the angular momentum of Pop III protostars reliably in simulations, extremely high resolution is required in order to distinguish between tight binaries with separations of a few astronomical units and single stars. One of the few sets of simulations that do have sufficient resolution are those presented by Greif et al. (2012). Stacy et al. (2013) analyzed the rotational properties of the protostars formed in these simulations and showed that most were rotating very rapidly, with rotational velocities between ~50% and 100% of the Keplerian velocity. However, these simulations cover only a very short period (~10 years) at the start of the evolution of the protostellar system and do not account for the effects of the magnetic field, which may provide strong magnetic braking on protostellar scales (Hirano & Bromm 2018). In view of these uncertainties, most studies of the impact of Pop III SNe on their surroundings have focused on the nonrotating case.

4.3.4.2. Impact of Population III supernovae. Pop III SNe affect their surroundings via both mechanical feedback (the injection of energy and momentum) and chemical feedback (the enrichment of the gas with metals). We first consider the role played by mechanical feedback. The gravitational binding energy of a minihalo with a mass close to $M_{\rm crit}$ is $\sim 10^{51}$ erg, which is comparable with the energy of a single CCSN. We would therefore expect a single CCSN to be able to eject a considerable fraction of the gas from a low-mass minihalo, provided that the SNR does not lose too much energy via radiative cooling.

In practice, the main factor that determines whether or not radiative cooling of the SNR is effective is the density of the region in which the SN explodes. If prior photoionization feedback

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has cleared away most of the gas, then cooling is ineffective and even a single SN can eject a large fraction of the gas from the halo (Kitayama & Yoshida 2005, Ritter et al. 2012, Jeon et al. 2014, Chiaki & Wise 2019). However, if prior photoionization feedback is ineffective, as would be the case for, e.g., a 10–15-$M_\odot$ CCSN, then little gas is ejected (Jeon et al. 2014, Chiaki & Wise 2019). Highly energetic SNe, such as HNe or PISNe, are less sensitive to the local environment and typically clear almost all gas out of the halo (Greif et al. 2010, Magg et al. 2022c). This picture changes in massive minihalos, owing to their larger gravitational binding energies. For minihalo masses larger than a few times $10^6\,M_\odot$, a single CCSN is no longer sufficient to eject the gas, although a single PISN remains effective (Chiaki & Wise 2019, Magg et al. 2022c).

The time taken for gas to become available again for star formation after the explosion of a Pop III SN, the so-called fallback or recovery time, varies considerably with halo mass and SN progenitor mass, ranging from a few megayears to 100 Myr or more. Smaller values are typically associated with lower-mass progenitors or higher-mass minihalos, but there is considerable scatter from halo to halo even for fixed progenitor mass. Magg et al. (2022c) argue that this is a consequence of the small-scale structure of the gas prior to the SN explosion. The greater the amount of gas located in dense clumps, the better it is able to survive the impact of the SNR and the shorter the resulting fallback time.

SNRs that successfully expel gas from their parent minihalo subsequently expand into the surrounding IGM, reaching final sizes of 500–1,500 pc by the time that they finish expanding (Greif et al. 2010, Jaacks et al. 2018), depending on their initial energy and the density of the surrounding gas. These sizes are large enough to reach neighboring minihalos. If these minihalos are already in the process of forming stars, then the interaction with the expanding SNR has little effect on them. It strips away the low-density gas in the outskirts of the minihalo but does not penetrate into the dense core (Chen et al. 2017). Minihalos that are still assembling their gas are more strongly affected, as in this case the expanding remnant can strip most of the gas from the halo, preventing it from cooling and collapsing (Sigward et al. 2005). However, these minihalos will already have been irradiated by photoionizing radiation from the SN progenitor (Jaacks et al. 2018), which has a similar suppressive effect, and so this direct mechanical feedback is generally considered to be of limited importance.

Turning to chemical feedback, we note that this proceeds in two qualitatively different ways: internal enrichment (i.e., enrichment of gas in the halo where the SN exploded) and external enrichment (i.e., enrichment of nearby halos that interact with the SNR). As we might expect from the discussion above, the relative importance of these mechanisms depends on how effectively SNe can drive gas and metals out of their host halos, with internal enrichment dominating in more massive halos and external enrichment dominating in lower-mass halos (Chiaki & Wise 2019). The likelihood of external enrichment occurring also depends on the local number density of minihalos capable of forming stars, as this determines how many are likely to be located within the region of the IGM enriched by the SN. Therefore, we would expect the importance of external enrichment to also depend on $M_{\rm crit}$, with this process becoming less important as $M_{\rm crit}$ increases. As an illustration, consider the studies of early metal enrichment by Hicks et al. (2021) and Magg et al. (2022c). Hicks et al. (2021) find that ~60% of the metal-enriched minihalos in their simulation gain their metals via external enrichment. However, these halos are small, with most having masses of $M < 10^6\,M_\odot$, and it is unclear how many will actually form stars. In comparison, Magg et al. (2022c) examine only the enrichment of halos that successfully form stars and find that in this case, only around 35% are externally enriched. Although these studies are a good start, more work along these lines is necessary to fully pin down the relative importance of external versus internal enrichment and how this depends on the local extragalactic environment, the strength of the LW background, etc.

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Finally, a quantity of great interest for the interpretation of observations of EMP stars in our own Galaxy (see Section 5.2.1) is the metallicity of the gas enriched by a single Pop III SN. This depends on the efficiency with which metals are mixed into the surrounding environment: The greater the amount of mixing, the lower the resulting metallicity. However, we can put an upper limit on this metallicity by assuming that metals only mix within the volume occupied by the SNR at the time that it finishes expanding. Magg et al. (2020) show that in this case, the minimum mass of gas into which the metals are mixed is given by

$$M_{\rm dil,min} = 1.9 \times 10^4 E_{51}^{0.96} n^{-0.11}\,{\rm M}_\odot, \qquad 23.$$

where E_{51} is the kinetic energy of the SN explosion in units of 10^{51} erg, and n is the number density of the gas in which the explosion takes place in units of $\rm cm^{-3}$. For explosions taking place within a preexisting HII region, $n^{-0.11} \sim 1$ and so a CCSN mixes its metals into a minimum of $2 \times 10^4\,{\rm M}_\odot$ of gas, whereas a PISN mixes its metals into a minimum of $2 \times 10^6\,{\rm M}_\odot$ of gas. The total mass of metals ejected by a Pop III CCSN ranges from around 0.1 to a few solar masses (Nomoto et al. 2006), and taking a value at the top end of this range then yields a maximum metallicity of $Z_{\rm max} \sim 0.01\,{\rm Z}_\odot$ for enrichment by a single CCSN. For a PISN, the mass of metals ejected is much larger, $\sim$100 $\rm M_\odot$ (Heger & Woosley 2002), but because this is mixed into a much larger gas mass, the resulting maximum metallicity is actually smaller, $Z_{\rm max} \sim 3 \times 10^{-3}\,{\rm Z}_\odot$. Magg et al. (2020, 2022c) show that in practice, individual Pop III SNe mix their metals into gas masses ranging from a few times this minimum value to as much as 10^4 times more. Large values typically correspond to cases of external enrichment, when a large mass of pristine gas associated with an existing minihalo is enriched by a small mass of metals from an external source. It should also be stressed that this mixing is generally not uniform, there can be substantial metallicity variations within the enriched volume (Ritter et al. 2015, Tarumi et al. 2020). Individual Pop III SNe can therefore produce metal-enriched gas with a broad range of metallicities, making it plausible that at least some of the EMP stars observed in the Milky Way will have formed from gas enriched by only a single Pop III SN (see also Section 5.2.1).

4.4. Transition to Population II Star Formation

Second-generation stars, sometimes termed early Pop II stars, formed from material that was enriched by metals from the first generation of stars. Unlike Pop III stars, which have not yet been directly detected (Section 5.1.1), members of the second generation have already been found in surveys looking for EMP stars in our Milky Way and neighboring satellite galaxies (Section 5.2.1). As discussed in Section 3.2.1, the chemical composition of the gas governs its thermodynamic response and, therefore, its fragmentation behavior under different environmental conditions. In turn, this determines the stellar mass spectrum (Section 3.2.2), which for genuine Pop III stars is predicted to be very different compared to what we find in the Universe today (Section 3.2.3). Unlike purely primordial gas, where cooling is largely provided by hydrogen (see **Figure 3*b***, subpanel *i*), metal-enriched gas has access to a much wider range of cooling mechanisms, allowing it to reach lower temperatures at various points in its evolution. This enhances the fragmentation of the gas and reduces the rate at which the fragments accrete additional material. Together, these effects are widely believed to lead to an IMF dominated by low-mass stars, similar to the present-day IMF (Kroupa 2002, Chabrier 2003). However, there remains the question of when and where this transition occurs, whether it is sudden or gradual, and what is the underlying physical cause for it. Indeed, there are two competing models based on two different low-metallicity cooling channels, each associated with a different critical metallicity.

The first model is based on atomic fine-structure lines from alpha elements such as carbon or oxygen. These start to dominate the cooling at number densities around $10^4\,{\rm cm^{-3}}$ and above a

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critical metallicity of about $10^{-3}\,Z_\odot$ (see, e.g., Chon et al. 2021). This leads to a markedly different evolution in the log T −log n diagram as indicated in **Figure 3*b***, subpanel *i*. The required metallicity (or, more precisely, the required carbon and oxygen abundances) can be estimated by comparing the cooling rate due to metals with the compressional heating rate at a characteristic temperature of 200 K and density of $10^4\,\mathrm{cm}^{-3}$ (see, e.g., Bromm & Loeb 2003, Santoro & Shull 2006), corresponding to the point at which H_2 cooling becomes inefficient in the Pop III case (see **Figure 3**). To quantify this, Frebel et al. (2007) introduced the transition discriminant, $D_{\mathrm{trans}} = \log_{10}(10^{[\mathrm{C/H}]} + 0.3 \times 10^{[\mathrm{O/H}]})$, and argued that low-mass stars can only form for metallicities of $D_{\mathrm{trans}} \gtrsim -3.5$.

The second line of reasoning considers dust cooling as the primary agent for fragmentation, which leads to a considerably lower critical metallicity in the range 10^{-5} to $10^{-6}\,Z_\odot$. This value was first estimated using analytical or simple one-zone numerical models that compared the strength of various cooling and heating processes in a metal-poor environment (e.g., Omukai et al. 2005; Schneider et al. 2006, 2012a; Chiaki et al. 2013a,b). However, it has subsequently received strong support from detailed 3D numerical simulations of dust cooling in dense, metal-poor gas (e.g., Tsuribe & Omukai 2006, 2008; Clark et al. 2008; Dopcke et al. 2013; Safranek-Shrader et al. 2014; Chiaki et al. 2016; Chiaki & Wise 2019; Chon et al. 2021; Chiaki & Yoshida 2022). Together, these models suggest that dust is the driving agent for the transition in the IMF. It occurs at a metallicity of about 10^{-5} solar and leads to an IMF peaking below $1\,M_\odot$ with a functional form similar to what is found in the Solar Neighborhood (Kroupa 2002, Chabrier 2003).

The existence of the star SDSS J1029151+172927 (Caffau et al. 2011a) provides strong evidence for the validity of the dust-induced fragmentation model (e.g., Klessen et al. 2012, Schneider et al. 2012b, Chiaki et al. 2014, Bovino et al. 2016) and is a challenge to models based on fine-structure cooling. SDSS J1029151+172927 is a truly primitive star in the constellation of Leo with a metallicity that falls well below the suggested critical value of D_{trans} for atomic fine-structure lines to be relevant. The star has elemental abundances in the range from 10^{-5} to 10^{-4} of the solar value for all of the elements measured in its spectrum (Caffau et al. 2012), setting it apart from other EMP stars that typically have enhanced CNO abundances despite being very iron poor (Beers & Christlieb 2005).

What is the physical origin of this behavior? Simple numerical experiments (Li et al. 2003) indicate that turbulent gas in a cooling regime (i.e., one in which T drops with increasing density and the effective adiabatic index is $\gamma_{\mathrm{eff}} < 1$) can fragment very efficiently to build up a cluster of low-mass stars. Conversely, star formation in a heating regime (T increasing with density, $\gamma_{\mathrm{eff}} > 1$) is biased toward forming very few and very massive objects. It has been posited (Larson 2005) that the transition between these regimes provides ideal conditions for efficient fragmentation and introduces a characteristic mass scale to the system. This mass scale is approximately the Jeans mass at the density and temperature of the transition point (Jappsen et al. 2005). It can be understood as a consequence of the fact that fragmentation in turbulent, self-gravitating gas often occurs in filaments, which are unstable against fragmentation for $\gamma_{\mathrm{eff}} < 1$ and stable for $\gamma_{\mathrm{eff}} > 1$ (Larson 1985), with isothermal gas constituting a critical case (Kawachi & Hanawa 1998). As all current models of the assembly of the first galaxies indicate that the gas develops a filamentary morphology, it is reasonable to assume that filament fragmentation dominates in the earliest galaxies just as it does in the local ISM of the Milky Way (Hacar et al. 2023).

The transition from $\gamma_{\mathrm{eff}} < 1$ to $\gamma_{\mathrm{eff}} > 1$ corresponds to a downward kink in the effective equation of state. Inspection of **Figure 3** reveals two sets of such kinks, one associated with atomic fine-structure line cooling (for metallicities $-3 \lesssim \log_{10} Z/Z_\odot \leq 0$) and one with dust cooling (in the range of $-5 \lesssim \log_{10} Z/Z_\odot \leq 0$). The associated characteristic masses are $\sim 100\,M_\odot$ for

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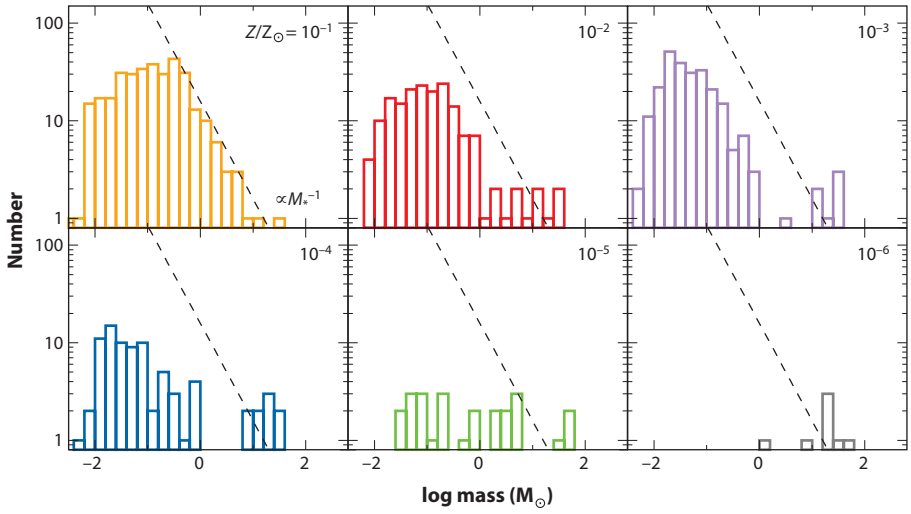

Figure 11

Illustration of the change of fragmentation behavior and resulting stellar mass spectrum for different metallicities, ranging from $10^{-6}\,Z_\odot$ to $10^{-1}\,Z_\odot$. For high metallicities the resulting IMF is bottom heavy with a peak below 1 $M_\odot$ and a power-law decline above, which is similar to the present-day IMF. At low Z the mass spectrum becomes logarithmically flat and top heavy similar to what we expect for the Population III IMF (cf. **Figure 6**). Figure adapted from Chon et al. (2021) with permission from MNRAS. Abbreviation: IMF, initial mass function.

fine-structure cooling and <1 $M_\odot$ for dust cooling. This implies that atomic fine-structure cooling is not able to produce the EMP low-mass stars we detect in stellar archeological surveys (see Section 5.2) in sufficiently large numbers, providing additional evidence for dust cooling being the dominant process driving the transition from the Pop III to the Pop II IMF.

To summarize the above, in **Figure 11** we show the stellar mass spectrum that results from systematically varying the metallicity of the star-forming gas (for further details, see Chon et al. 2021). Although similar caveats apply here as for our previous discussion of the Pop III IMF, we nevertheless see a clear influence of metallicity. Values close to solar lead to a bottom-heavy IMF which peaks around 0.2–0.3 $M_\odot$ with a power-law drop toward larger masses, similar to the present-day values (Kroupa 2002, Chabrier 2003). As the metallicity gets smaller, deviations from the present-day IMF become more pronounced and the mass spectrum becomes logarithmically flat and top heavy, which is more comparable to the Pop III IMF.

5. OBSERVATIONAL PROBES OF Pop III STARS

Characterizing the properties of Pop III stars is difficult, as most of what we know about them comes from theoretical models and numerical simulations rather than observational constraints, which are hard to obtain and highly indirect at best. In this section, we provide an overview of the available data from the high-redshift and the present-day Universe.

5.1. High-Redshift Universe

There are several different approaches that we can take when trying to use observations of the high-redshift Universe to constrain the properties of Pop III stars. First, we can try to detect the stars themselves. Second, we can observe the bright explosive transients that the most massive

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Pop III stars produce at the end of their lives (SNe and gamma-ray bursts). Finally, we can look for larger-scale signatures of the energetic feedback produced by massive Pop III stars, particularly through the imprint they leave on the cosmic 21-cm background.

5.1.1. Direct detection of Population III stars. Many of our uncertainties regarding the formation of Pop III stars could be resolved if we could observe them directly in the high-redshift Universe. The possibility of doing so with current or near-future facilities has been investigated by a number of authors (see, e.g., Oh et al. 2001; Scannapieco et al. 2003; Greif et al. 2009; Zackrisson et al. 2011, 2012; Rydberg et al. 2013; Mas-Ribas et al. 2016; Riaz et al. 2022). Unfortunately, the general conclusion of these studies is that this is not possible, at least if we are interested in Pop III stars with masses <1,000 $M_\odot$ forming in minihalos or atomic cooling halos with efficiencies similar to those predicted by numerical simulations. A recent calculation by Schauer et al. (2020) highlights the problem. They show that a single 1,000-$M_\odot$ Pop III star forming at a redshift of $z > 10$ produces a flux in the NIRCAM wide-field filters on JWST of between 5×10^{-4} and 2×10^{-3} nJy, considering only the stellar emission. Accounting for nebular emission increases these numbers by about a factor of ten, but they are still far below the limiting values of a few nanojanskys reachable with deep JWST imaging (see, e.g., Merlin et al. 2022), although they would be in reach of a hypothetical 100-m extraterrestrial telescope (Schauer et al. 2020).

In order to be detectable by JWST, our source therefore needs to be about 100 times brighter than we expect for a single very massive Pop III star. One way to reach the necessary brightness would be to have a cluster of such stars. However, the required stellar mass—around 10^5 $M_\odot$ in massive Pop III stars (Zackrisson et al. 2011)—does not seem achievable in a minihalo, given what we know about Pop III star formation. For instance, simply building up this much mass in a central cluster given the typical mass inflow rates we measure in simulations would require ~10 Myr, but feedback from the assembling cluster disrupts the inflow long before this. Furthermore, the required star-formation efficiencies are unreasonably large; for instance, a minihalo with a mass close to $M_{\rm crit}$ would have to convert close to 100% of its baryonic mass into stars. Clusters of the required size could conceivably build up in more massive galaxies, but the availability of sufficient metal-free gas in these systems is unclear. Models for the evolution of the Pop III SFRD agree that some level of Pop III star formation persists down to $z \sim 5$–6 (see **Figure 2**), but disagree greatly on the average mass in young Pop III stars per galaxy. For example, Xu et al. (2016), Mebane et al. (2018), Liu & Bromm (2020b), and Skinner & Wise (2020) argue that close to reionization, the typical mass in young Pop III stars per galaxy is $\sim 10^3$ $M_\odot$, whereas Pallottini et al. (2014) and Sarmento et al. (2018) find much larger values of $\sim 10^5$ $M_\odot$. This difference stems from differences in how Pop III star formation in minihalos is treated in the models. If the process is strongly suppressed by LW radiation, baryonic streaming, or the impact of limited numerical resolution, then there is less preenrichment accreted by more massive galaxies, making it more likely to find large clusters of Pop III stars in these systems. In contrast, if Pop III star formation in minihalos is barely affected by early feedback, as in, e.g., Skinner & Wise (2020), then there is greater preenrichment of the gas in massive galaxies, resulting in smaller Pop III cluster masses. Intriguingly, this suggests that a nondetection by JWST of any Pop III-dominated high-redshift galaxies may allow us to constrain the efficiency of star formation in minihalos.

Another way to reach the brightness necessary for JWST to detect small clusters of massive Pop III stars is gravitational lensing. For the example discussed above, we would need a magnification factor of around 100 in order to increase the brightness sufficiently. To detect individual Pop III stars at high redshift, even higher magnifications would be required. Rydberg et al. (2013) estimate that a magnification of around 3,000 is required for JWST to detect a lensed 60-$M_\odot$ star at $z \sim 10$. Such high magnifications are achievable close to lensing caustics (Windhorst et al. 2018,

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Diego 2019), but the disadvantage is that only a very small fraction of the high-redshift Universe is magnified by such a large amount. The chance of actually detecting a highly lensed Pop III star therefore depends on the number of massive Pop III stars present in the high-redshift Universe and, hence, on the Pop III SFRD. Unfortunately, even for values of this at the upper range of predictions from the literature (see Section 3.1.3), the probability of actually detecting a highly lensed Pop III star or star cluster is small (Rydberg et al. 2013, Diego 2019). Despite this, two possible candidate objects have recently been reported in the literature: a highly lensed star cluster at $z = 6.629$ (Vanzella et al. 2020) and *Earendel*, an extremely magnified single star at $z = 6.2$ (Welch et al. 2022), which Schauer et al. (2022) argue has a nonnegligible chance of being a Pop III star. Whether either of these objects is truly primordial is currently unclear but is likely to be resolved by JWST in the near future.

Finally, if some fraction of Pop III stars are supermassive, with masses of $M \sim 10^5\,\mathrm{M}_\odot$ and luminosities of $L \sim 10^9\,\mathrm{L}_\odot$, then these may be directly detectable by JWST with no or minimal lensing (Surace et al. 2018, 2019; Vikaeus et al. 2022). They may also be detectable by *Euclid* and the *Nancy Grace Roman Space Telescope* (RST), given magnifications of 10–1,000 (Vikaeus et al. 2022). However, we expect these stars to be rare objects (see discussion in Section 3.4), which strongly limits the number we expect to detect in a typical survey. For example, Vikaeus et al. (2022) show that with reasonable survey parameters RST only detects ~10 SMSs, even if their number density at high redshift is close to the predicted upper limits, whereas *Euclid* is unlikely to detect any. However, JWST should detect ~30 or more in the redshift range of $7 < z < 10$.

It is also important to note that detecting Pop III stars in the early Universe is only part of the problem. Equally important is our ability to distinguish them from other high-redshift objects. In the observer's frame, Pop III stars have very red colors owing to their high redshifts and the almost complete absorption of radiation shortward of Lyman-α by the neutral IGM. This is true regardless of their rest-frame colors but is exacerbated in cases in which pre-MS Pop III stars or highly inflated massive stars contribute significantly to the observed flux (Mitani et al. 2019, Woods et al. 2021c). However, Pop II-dominated galaxies at the same redshift are also very red, as are dusty starbursts at lower redshifts (Naidu et al. 2022). Pop III-dominated systems can potentially be distinguished from Pop II-dominated systems based on their broadband colors, but doing this purely with the NIRCAM filter set requires an almost complete absence of nebular emission (i.e., a very high escape fraction of ionizing photons, $f_{\mathrm{esc}} > 0.95$; see Zackrisson et al. 2011). For galaxies with lower escape fractions, detection in the MIRI 560W and 770W filters is also necessary for distinguishing between Pop III and Pop II stars, limiting this method to only the brightest sources (Zackrisson et al. 2011, Trussler et al. 2022).

A more promising route for distinguishing Pop III stars from other high-redshift objects involves spectroscopy. Massive Pop III stars are unusually hot compared to metal-enriched stars (Schaerer 2002) and, hence, produce a much larger flux of photons with energies above 54 eV that are capable of ionizing He^+ to He^{++}. It was therefore quickly realized that the recombination lines of HeII, particularly the 1640-Å line, could act as an important diagnostic of Pop III star formation (Oh et al. 2001, Tumlinson et al. 2001, Schaerer 2002). Unfortunately, detection of this line by itself does not prove that the source is a Pop III star, as Wolf–Rayet stars, X-ray binaries, and black holes can also produce strong HeII emission (see, e.g., Erb et al. 2010, Schaerer et al. 2019). A salutary example is provided by the $z = 6.6$ Lyman-α emitter CR7 (Sobral et al. 2015). It shows a strong HeII 1640-Å emission line that was interpreted by some authors as being due to Pop III star formation (e.g., Pallottini et al. 2015, Visbal et al. 2016). However, subsequent observations detected [OIII] emission from this source (Bowler et al. 2017), which is inconsistent with this explanation, and it is now thought that CR7 is either a low-luminosity AGN or a metal-poor starburst. A Pop III origin for HeII 1640-Å emission becomes more plausible if there is no

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associated emission from highly ionized metals, but even in this case a Pop II origin cannot be excluded (Katz et al. 2022). Altogether, the equivalent width of HeII recombination lines appears to be a promising diagnostic parameter, with Pop III stars producing larger widths than other astrophysical sources (Nakajima & Maiolino 2022), although the requirement to measure it accurately may again restrict the use to only the brightest sources.

5.1.2. Population III transients: supernovae and gamma-ray bursts. The high peak brightness of Pop III CCSNe and PISNe potentially make them much easier targets to observe than their massive star progenitors. The predicted peak luminosity of a Pop III CCSN can reach values as high as a few times 10^{11} $L_\odot$ (Whalen et al. 2013a), around 10^4 times greater than the luminosity of the single 1,000-$M_\odot$ star. Pop III PISNe are even brighter, with peak luminosities a factor of ten or more greater than for CCSNe (Whalen et al. 2013b). Although these peak luminosities persist for only brief periods, Pop III SNe remain brighter than their progenitors over a period of weeks to months.

In a series of papers, Whalen and collaborators have examined the detectability of Pop III CCSNe, HNe, PISNe, and pulsational PISNe by JWST and other future facilities (Whalen et al. 2013a,b, 2014; Smidt et al. 2014, 2015). They show that Pop III PISNe are bright enough to detect with JWST at redshifts of $7 < z < 15$ for hundreds of days. The brighter explosions may also be marginally detectable with the RST, but are likely too faint to detect with *Euclid* or in the Pan-STARRS (Panoramic Survey Telescope & Rapid Response System) or LSST (Legacy Survey of Space and Time on the Rubin Observatory) sky surveys. Similar results are found for pulsational PISNe and HNe. CCSNe are harder to detect: Some explosion models are bright enough to be detectable with JWST at high redshift but others are not, and all are likely too faint to detect with the RST.

Unfortunately, although many Pop III SNe may be bright enough for JWST to detect, we expect them to be rare because of their short duration, and so the expected number of detections is small. The Pop III SNe rate depends on the star-formation rate and IMF of Pop III stars and, hence, is subject to the considerable uncertainties in both of these quantities that we have discussed previously. However, several recent studies broadly agree that we should expect around $3–7 \times 10^{-4}$ Pop III PISNe per JWST NIRCAM field of view per year (Hummel et al. 2012, Johnson et al. 2013, Magg et al. 2016, Hartwig et al. 2018a). The numbers for CCSNe are larger, but not by more than an order of magnitude (Magg et al. 2016). The chance of detecting any Pop III SNe in a dedicated survey is therefore tiny. Serendipitous detections remain plausible, although unambiguously identifying the detected objects as Pop III SNe without multiepoch imaging may be problematic (Hartwig et al. 2018a). At redshifts close to the epoch of reionization, confusion with SNe produced by low-metallicity Pop II stars will also be a major issue (Moriya et al. 2022).

The other transients related to Pop III star formation that are detectable at cosmological distances are long-duration GRBs. The extremely high luminosities associated with these events make them readily visible even at high redshifts, as demonstrated by GRB 090423, which has a spectroscopically confirmed redshift of $z \approx 8.2$ (Tanvir et al. 2009) and GRB 090429B, which has a photometric redshift of $z \approx 9.4$ (Cucchiara et al. 2011). Current gamma-ray telescopes such as the *Swift Observatory* (Gehrels et al. 2004) are unlikely to detect GRBs at redshifts much higher than $z \sim 9$ because of limited sensitivity and, hence, are mostly sensitive to GRBs produced by metal-enriched stars (Ghirlanda et al. 2015). However, future gamma-ray facilities such as THESEUS (*Transient High-Energy Sky and Early Universe Surveyor*) will expand the redshift range in which GRBs and their afterglows can be readily detected up to $z \sim 20$ (Amati et al. 2018), allowing us to probe redshifts at which Pop III sources will dominate (see also Wang et al. 2012). In view of this, it is clearly of interest to be able to make predictions for the number of Pop III GRBs

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that we expect these future facilities to detect. Unfortunately, this turns out to be even less well constrained than the number of Pop III SNe we expect at high redshift. As well as uncertainties due to the Pop III SFRD and IMF, the number of GRBs that we expect to detect also depends on several other highly uncertain numbers (Bromm & Loeb 2006): the fraction of massive Pop III stars that are found in close binary systems, the fraction of these stars that are rapidly rotating, and the beaming factor (i.e., the probability that a randomly selected GRB happens to beam its radiation in our direction). Estimates of the number of Pop III GRBs that we should detect given a perfect telescope therefore range from values as high $\sim$100 year^{-1} (Kinugawa et al. 2019, Lazar & Bromm 2022) to ones as small as 0.1 year^{-1} (Toma et al. 2016).

Spin temperature, T_s: effective temperature that would produce the observed ratio of parallel to antiparallel spins if the atomic hydrogen gas were in thermal equilibrium

5.1.3. 21-cm global signal. The ground state of atomic hydrogen exhibits a hyperfine splitting owing to the interaction between the proton and electron spins, with an energy difference of 5.9×10^{-6} eV (corresponding to $\lambda \approx 21$ cm) between the two states. The radiative transition between these states—the 21-cm line—is frequently used to determine the properties of atomic hydrogen from the local Universe to very high redshifts. The global (i.e., sky-averaged) 21-cm signal is an important probe of the Universe at the redshifts when the first stars formed. When averaged on very large scales, the contribution of density fluctuations cancels out, and the strength of the line is determined solely by the spin temperature T_s of the transition, yielding an emission line if this is larger than the CMB temperature T_{CMB}, and an absorption line if it is smaller. The evolution of T_s is controlled by three physical processes: collisions with H, H^+, and e^-, the absorption and emission of 21-cm photons, and the Wouthuysen–Field effect (i.e., hyperfine transitions brought about by the absorption and emission of Lyman-α photons; see Wouthuysen 1952, Field 1958).

At redshifts immediately prior to the formation of the first stars, the interaction with the CMB dominates, and $T_s \approx T_{CMB}$. However, once Pop III stars start to form, the Lyman-α photons that they produce allow the Wouthuysen–Field effect to become dominant, driving T_s toward the kinetic temperature of the gas. The timing of this transition therefore lets us learn something about when (and how quickly) Pop III stars started to form, and the information that it provides us on the temperature of the gas constrains the rate at which the high-redshift IGM is heated by X-ray sources (e.g., Fialkov et al. 2014, Pacucci et al. 2014), its degree of ionization (which becomes more important at lower redshift; Park et al. 2020), and possibly also the coupling between dark matter and gas (e.g., Barkana 2018, Muñoz et al. 2018, Liu et al. 2019). A measurement of the global 21-cm signal could also in principle test the existence of any radio background in addition to the CMB (e.g., Ewall-Wice et al. 2018, Feng & Holder 2018, Fialkov & Barkana 2019, Reis et al. 2020). Measurement of the high-z 21-cm line therefore provides stringent constraints on the early evolution of the cosmic star-formation history and the properties of the first stars (Madau & Dickinson 2014), as reviewed by Furlanetto et al. (2006), Barkana & Loeb (2001), Fan et al. (2006a), Morales & Wyithe (2010), Pritchard & Loeb (2012), or Mesinger (2019).

Experiments measuring the global signal at high redshifts include EDGES (Experiment to Detect the Global EoR Signature; Bowman et al. 2013), LEDA (Large-Aperture Experiment to Detect the Dark Age; Price et al. 2018), SARAS (Shaped Antenna measurement of the background RAdio Spectrum 2; Singh et al. 2018), PRIZM (Probing Radio Intensity at high-Z from Marion; Philip et al. 2019), or Reach (Radio Experiment for the Analysis of Cosmic Hydrogen; de Lera Acedo et al. 2022). The first tentative detection of this signal was reported by the EDGES collaboration (Bowman et al. 2018). Their signal is much stronger than expected, and efforts to explain this have led to considerable theoretical speculation but few firm conclusions. In addition, we note that the true nature of this signal is highly debated (e.g., see Hills et al. 2018, Sims & Pober 2020). If it is of cosmological origin it represents one of our first direct constraints on the star-formation process at a redshift of $z \sim 17$ as reflected by the impact of Lyman-α coupling and X-ray heating.

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The characteristics of early star formation can influence the global 21-cm signal in a variety of additional ways. For example, the recovery time (i.e., the average duration for metal-enriched material, which has been expelled from the Pop III star-forming halo, to mix with the ambient material and recollapse to form Pop II stars; see Section 4.3.4.2) can influence the observable 21-cm signal. Mirocha et al. (2018) and Magg et al. (2022b) find that short recovery times, resulting from weak SN feedback, possibly due to Pop III star formation occurring in more massive halos or with low efficiency, result in a noticeably steeper global 21-cm signal than models with long recovery times from strong feedback, characteristic of scenarios in which stars build up in low-mass halos or form in groups leading to mechanical energy and momentum input from multiple SNe. The global 21-cm signal is also sensitive to the Pop III IMF. In a study focusing on the impact of Lyman-α radiation at high redshift and neglecting ionization and X-ray emission, Gessey-Jones et al. (2022) show that current and future observations can probe the characteristic stellar mass if the IMF is dominated by stars lighter than $\sim 20\,M_\odot$. However, the expected signatures are relatively weak and comparable with the instrumental precision. If the Pop III IMF is dominated by very massive stars, we can no longer discriminate between different IMF models. This may change once calculations including X-rays have been performed, which are thought to have a different dependence on the IMF compared to the Lyman-α emission. Detailed measurements of the 21-cm emission at high redshifts, once available, will open up many interesting pathways to study the birth of Pop III stars and the transition to Pop II star formation.

5.2. Low-Redshift Universe

The most stringent constraints on the properties and mass distribution of Pop III stars are likely to come from the study of EMP stars in the Local Group. These stars have a metallicity of $[\mathrm{Fe/H}] < -3$ as expressed by the iron abundance relative to the solar value (for a review, see Beers & Christlieb 2005). A common approach is to use large-area photometric or low-resolution spectroscopic surveys[6] for a first estimate of the stellar metallicity (e.g., Ludwig et al. 2008, Caffau et al. 2011b). This information can then be used to put together a list of objects for high-resolution spectroscopic follow-up observations. Good targets are MS turnoff stars, because they are numerous and bright, and they have chemical abundances[7] that are essentially unaltered since the time they were formed (Pinsonneault 1997). The resulting stellar archeological surveys, such as those aiming at the Galactic halo and/or bulge (Christlieb et al. 2008, Schörck et al. 2009, Frebel 2010, Salvadori et al. 2010, Caffau et al. 2013, Starkenburg et al. 2017, Arentsen et al. 2020, Ishigaki et al. 2021, Li et al. 2022) or nearby satellite dwarf galaxies (e.g., Koch et al. 2013; Kirby et al. 2015; Salvadori et al. 2015; Ji et al. 2016a,b, 2023; Roederer et al. 2016), can contribute to our knowledge of primordial stars in two key ways.

5.2.1. Galactic archaeology and the high-mass end of the Population III initial mass function. We can use the abundance pattern determined in EMP stars to infer the properties of the progenitor stars that provided the observed chemical enrichment. It is common practice to employ stellar evolution calculations combined with explosive nucleosynthesis models for this comparison, for both individual stars (e.g., Ishigaki et al. 2014a; Bessell et al. 2015; Placco et al. 2016; Aguado et al. 2018a, 2019) as well as large samples (Fraser et al. 2017, Ishigaki et al. 2018). This is facilitated by libraries of modeled SN yields (Nomoto et al. 2006, Umeda & Nomoto 2008,

[6]For example, the Sloan Digital Sky Survey provides spectral data for several 10^5 stars (for DR12, see García Pérez et al. 2016), and Gaia for more than 10^8 sources (for DR3, see Gaia Collab. et al. 2022).

[7]We adopt the notation $[\mathrm{X/H}] = \log_{10}(m_\mathrm{X}/m_\mathrm{H}) - \log_{10}(m_{\mathrm{X},\odot}/m_{\mathrm{H},\odot})$, where m_X and m_H are the mass abundances of any element X and hydrogen, and $m_{\mathrm{X},\odot}$ and $m_{\mathrm{H},\odot}$ are the corresponding solar abundances.

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Heger & Woosley 2010, Ishigaki et al. 2018), which typically depend on the stellar mass of the exploding star, the explosion energy, and one or several parameters that quantify the mixing-and-fallback process. The latter cannot be simulated self-consistently in 1D SN models and in principle require detailed 3D radiation–hydrodynamic calculations of the star-forming halo for determination (Magg et al. 2022c). An additional constraint on this process is provided by the fact that the SN yields need to be diluted with metal-free gas in the surrounding of the exploding star in order to match the absolute metallicity of the observed star (Magg et al. 2019).

The observational data indicate that most EMP stars are rich in carbon in relative terms (Beers & Christlieb 2005, Frebel & Norris 2015). Of specific interest among these carbon-enhanced metal-poor (CEMP) stars are the so-called CEMP-no stars. Despite an excess of carbon relative to iron of more than [C/Fe] = 1, they show no enhancement in neutron-capture elements; i.e., they have [Ba/Fe] < 0 (Frebel et al. 2005, Komiya et al. 2007, Norris et al. 2013, Keller et al. 2014, Aguado et al. 2018b, Nordlander et al. 2019). The origin of the elemental abundance patterns in these stars is one of the key questions of early chemical evolution. Their dynamics are inconsistent with them having gained their metals via mass transfer from a binary companion (Zepeda et al. 2023). Umeda & Nomoto (2002) propose that the abundance pattern of CEMP-no stars is the fingerprint of a so-called mixing-and-fallback or faint SN (see also Ishigaki et al. 2014b, Chan et al. 2020). Such events have relatively low explosion energies, and consequently the inner layers of the star, containing in particular iron, are suggested to fall back on the compact remnant, such that only the outer layers, containing carbon and other light elements, are ejected and become mixed with ambient material.

Assuming that the oldest and most metal-poor stars in the Galaxy have been supplied with heavy elements by only one or at most two SN explosions—a common assumption but one that may not be true for all EMP stars (Hartwig et al. 2018b, 2019; Welsh et al. 2021)—one finds that the relative abundances of heavy elements are most consistent with CCSNe from Pop III stars in the mass range of 20–40 $M_\odot$ (e.g., Frebel et al. 2005; Iwamoto et al. 2005; Lai et al. 2008; Joggerst et al. 2009, 2010; Caffau et al. 2011a, 2012; Norris et al. 2013; Cooke & Madau 2014; Keller et al. 2014; Bonifacio et al. 2015), although a few examples exist of stars that are more consistent with enrichment by HNe (Placco et al. 2021, Skúladóttir et al. 2021) or PISNe (Aoki et al. 2014, Salvadori et al. 2019). The conclusion that most Pop III SNe were CCSNe is consistent with the work of Mapelli et al. (2006), who compute the expected density of intermediate-mass black holes in the Galaxy by assuming they are the relics of higher-mass Pop III stars. This allows them to derive an upper limit from the nondetection of ultraluminous X-ray sources in the Milky Way. The approach has been extended by de Bennassuti et al. (2014, 2017), who also include the Galactic metallicity function and the abundances of EMP stars. Again these data appear inconsistent with PISNe being important (although see Salvadori et al. 2019 for a contrary view). Taken together, these studies suggest that most Pop III stars forming through the standard pathway outlined in Section 3.2 are not likely to reach masses significantly above $\sim$40 $M_\odot$, with more massive objects being extremely rare and possibly being the result of collisions in very dense clusters (see Section 5.3).

Finally, we note that complementary information on the metal yields produced by the first generation of stars is potentially provided by observations of extremely metal-poor foreground absorption systems in the spectra of distant quasars. The most metal poor of these systems have metallicities comparable with EMP stars (e.g., Cooke et al. 2017, Robert et al. 2022, Welsh et al. 2022). If we assume that each of these systems has been enriched by either a single Pop III SN or a small number of such SNe, then we can use the observed metal abundances to infer their properties. In general, the results we obtain from this procedure are consistent with those coming from EMP stars. Most of the systems show elemental abundances consistent with enrichment

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primarily by CCSNe (Welsh et al. 2019), although a recent example has been found of a cloud in the broad-line region of a $z = 7.6$ quasar that has an abundance pattern that indicates a PISN origin (Yoshii et al. 2022). This supports the idea that most Pop III SNe are CCSNe, with PISNe occurring only rarely.

5.2.2. Potential for direct detection of Population III stars and the low-mass end of the initial mass function. Stellar evolution models (e.g., Kippenhahn et al. 2012, Maeder & Meynet 2012) tell us that low-mass Pop III stars with $M < 0.8\,M_\odot$ must have survived until the present day. Furthermore, models of the surface pollution of Pop III stars by metals from the ISM suggest that this is not an important effect and that Pop III stars around at the present day would still be spectroscopically identifiable as such (Frebel et al. 2009, Tanaka et al. 2017). Therefore, if these stars exist, there is a chance to directly detect some of them in current or future stellar archeological surveys. One can use semianalytic models to predict the most likely locations to find these Pop III survivors. Whereas the predicted numbers of survivors are highest in the Galaxy, specifically in the bulge, they are very difficult to find there because of the presence of so many higher-metallicity stars. It is therefore better to search in environments that did not have continuous star formation over secular timescales but instead only experienced an initial burst of star formation at high redshift and then remained quenched during subsequent cosmic evolution. Consequently, the lowest-mass satellites of the Milky Way are expected to contain the largest fraction of Pop III survivors compared to their Pop II stellar content, and so they are the most promising targets for future attempts to find Pop III survivors with high-resolution, high signal-to-noise spectroscopic observations (Magg et al. 2018). **Figure 12** shows predictions of the expected fraction of genuine low-mass Pop III stars in Local Group galaxies together with their total stellar mass. We see that the lowest-mass satellites are the best places to look for these stars, although it is possible that some of these systems are too small to contain any Pop III survivors.

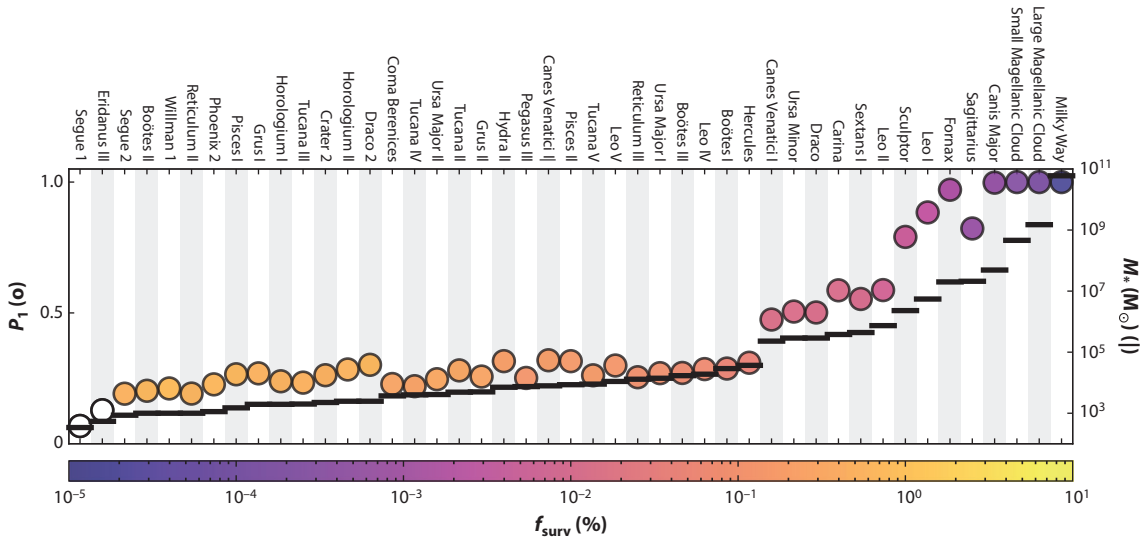

Figure 12

Properties of the Milky Way and its satellite galaxies. The horizontal line in each column indicates the stellar mass as inferred from observations. The filled circles indicate the probability that the systems contains at least one low-mass Population III survivor, with the color encoding the median fraction. The figure is based on data from Magg et al. (2018).

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Even nondetections allow us to put stringent limits on the low-mass end of the Pop III IMF. For example, Hartwig et al. (2015) estimate the expected numbers of low-mass Pop III stars in the Galactic halo based on semianalytic models of the early star-formation history in Milky Way–like halos. They conclude that if no such object is found in surveys with sample sizes of 4 million stars, then we can exclude the existence of low-mass Pop III stars with masses below $0.8\,M_\odot$ with high significance. On similar grounds, Salvadori et al. (2007, 2010) develop a detailed model of the metallicity distribution function of metal-poor stars in the Galactic halo and suggest that Pop III stars should be more massive than $0.9\,M_\odot$ when matching their predictions with the observational data. A different conclusion was reached by Tumlinson (2006, 2010), who argues that the current abundance measurements are not precise enough to distinguish between different models of the primordial IMF based on studying the chemical evolution during the early buildup of the Galaxy. However, he also suggests that characteristic masses of a few tens of solar masses provide a better fit to the available data than masses of $100\,M_\odot$ or above, which is consistent with the SN yields mentioned above. Finally, Rossi et al. (2021) argue that our failure to find any Pop III stars in nearby ultrafaint dwarf galaxies already allows us to exclude Pop III stars with $M < 0.8\,M_\odot$. Altogether, the prospects of finding surviving low-mass first stars in our immediate neighborhood are highly exciting, but at present no firm conclusions about the low-mass end of the Pop III IMF can be drawn. This may change with the availability of the new generations of multiobject spectrographs, such as 4MOST (4-metre Multi-Object Spectroscopic Telescope) or WEAVE (William Herschel Telescope Enhanced Area Velocity Explorer; e.g., Dalton et al. 2012, Feltzing 2016, Feltzing et al. 2018), which will allow for galactic archeological surveys of unprecedented size and depth that can help to resolve this issue.

5.3. Gravitational Waves

The detection of gravitational waves with extremely precise interferometers [advanced LIGO (Laser Interferometer Gravitation-Wave Observatory; Aasi et al. 2015), advanced Virgo (Acernese et al. 2015), or KAGRA (Kamioka Gravitational Wave Detector; Aso et al. 2013), all of which are currently operational, and the upcoming LISA (*Laser Interferometer Space Antenna*; Amaro-Seoane et al. 2023), DECIGO (*DECi-hertz Interferometer Gravitational wave Observatory*; Kawamura et al. 2011), and *TianQin* (Luo et al. 2016) experiments in space or the Einstein Telescope on the ground (Punturo et al. 2010)] opens up a new observational window to the Universe. The signals detected so far come from the gravitational inspiraling and merging of two stellar-mass black holes, two neutron stars, or pairs of a black hole and a neutron star. Whereas LIGO/Virgo/KAGRA can detect the mergers of compact objects with masses up to several hundred solar masses out to $z \sim 1$ (e.g., Abbott et al. 2019), the planned observatories (such as LISA, *TianQin*, and DECIGO, or the Einstein Telescope) have the potential to detect considerably more massive objects out to greater redshifts.

There are two primary ways to infer information about primordial stars from these data. On one hand, we can look at the properties of individual measurements, which gives us information about the total mass of the system, the mass ratio of the two constituents, and their spin. This provides constraints on the mass spectrum and multiplicity of Pop III stars and on possible formation and evolution pathways. On the other hand, we can study the rate of these events (typically normalized to a cosmic volume of 1 Gpc^3) as a function of redshift and compare the results with our understanding of the cosmic star-formation history. However, the astrophysical interpretation of these data is highly challenging for several reasons. First, the detected gravitational wave signal from the merger of two compact objects can occur many gigayears after the formation of the original stellar binary system. To link these two events requires complex binary evolution calculations

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that reproduce the correct stellar structure in the MS and post-MS phases and that reliably model mass transfer and the change of orbital parameters over secular timescales (see, e.g., Belczynski et al. 2002, 2016; de Mink & Mandel 2016; Eldridge & Stanway 2016; Mandel & de Mink 2016; Marchant et al. 2016, 2021; Mapelli et al. 2017; Stevenson et al. 2017; Spera et al. 2019). Second, to explain the current data, the mergers of stars in a dense cluster environment may be necessary. The corresponding stellar dynamical processes contribute additional stochasticity and uncertainty to the problem (e.g., Portegies Zwart & McMillan 2002, Portegies Zwart et al. 2022). And finally, the signal coming from genuine Pop III stars is much weaker than the rates expected from metal-enriched binaries, simply because the total number of metal-free stars is many orders of magnitude smaller, which can be seen, for example, by comparing the corresponding cosmic star-formation rate densities in **Figure 2** and which cannot be compensated by differences in the IMF (**Figure 11**).

The combined data of the first three observing runs of LIGO, Virgo, and KAGRA (Abbott et al. 2021, 2023) demonstrate that the mass spectrum inferred for the primary and secondary merger components is very wide and contains structure beyond a simple power law with a sharp high-mass cutoff. The mass distribution of the primary shows a first peak at $\sim$10 $M_\odot$ and a secondary one at 30–35 $M_\odot$. Although the majority of the systems detected have primary masses smaller than 45 $M_\odot$, the mass distribution clearly extends beyond $\sim$65 $M_\odot$ and reaches into the so-called pair-instability gap, at least for metal-rich stars (e.g., Farmer et al. 2019, Woosley & Heger 2021; see also Section 4.3.4). Pair-instability and pulsational PISNe lead to the complete disruption of the star and leave no remnant behind. Because metal-rich stars experience considerable mass loss due to strong winds or pulsational instabilities prior to the explosion, we expect that no black hole in the mass range between about 50 $M_\odot$ and 130 $M_\odot$ should form, whereas for zero-metallicity stars, which experience no mass loss, the pair-instability gap covers masses from $\sim$90 $M_\odot$ to $\sim$260 $M_\odot$ (however, with considerable uncertainties, see, e.g., Heger & Woosley 2002, Woosley & Heger 2015, Farmer et al. 2020, Costa et al. 2021, Vink et al. 2021).

As we expect that most of the observed gravitational wave detections come from the remnants of metal-enriched rather than zero-metallicity stars (e.g., Hartwig et al. 2016, Belczynski et al. 2017, Tanikawa et al. 2022), these findings cast doubts on the widespread occurrence of PISNe and it has been argued that stellar collisions in a dense cluster environment provide an alternative pathway to produce the high-mass end of the observed black hole mass spectrum (e.g., Portegies Zwart & McMillan 2000, 2002; Portegies Zwart et al. 2004; Giersz et al. 2015; Mapelli 2016; Di Carlo et al. 2019, 2021; Liu & Bromm 2020a; Dall'Amico et al. 2021; Costa et al. 2022; Wang et al. 2022). At the low-mass end of the distribution, the data indicate a lack of low-mass black holes below $\sim$6 $M_\odot$ with high significance. This is greater than the masses reported for black hole candidates in Galactic binaries (reaching down to $\sim$3 $M_\odot$; see Thompson et al. 2019) and potentially hints toward a more top-heavy IMF of the progenitor stars. Most systems have mass ratios q close to unity, but there are some for which $q < 1$ is inferred with high significance.

Another feature of these reported gravitational wave detections is that about one third of the systems exhibit spins that are possibly inclined with respect to the orbital angular momentum and that show signs of precession (see also Callister et al. 2022). This provides additional support for a collisional origin of these systems. The inferred merger rates for binary black holes have a 90% credibility interval of 18–44 $\mathrm{Gpc}^{-3}\ \mathrm{year}^{-1}$; those for binary neutron stars lie in the range of 10 to 1,700 $\mathrm{Gpc}^{-3}\ \mathrm{year}^{-1}$ (Abbott et al. 2023). These values are still highly uncertain, but the precision will increase with time as the number of detected gravitational wave events continues to go up. The data furthermore indicate a merger rate that grows with redshift as $(1 + z)^\kappa$, with κ only marginally constrained in the range from 1.1 to 4.6, at least for $z \lesssim 1$. Theoretical predictions of gravitational wave event rates, specifically from the remnants of Pop III stars (Kinugawa et al.

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2014; Hartwig et al. 2016; Belczynski et al. 2017, 2022; Schneider et al. 2017; Dayal et al. 2019; Neijssel et al. 2019; Tang et al. 2020; Ng et al. 2021; Tanikawa et al. 2021, 2022), are roughly consistent with these numbers, but more stringent constraints clearly require larger statistics (for a comprehensive overview of this aspect, see Mandel & Broekgaarden 2022).

SUMMARY POINTS

1. The formation of the first stars begins at redshifts of $z \sim 30$ and is likely to continue to $z \sim 5$, dominating the cosmic star-formation rate density until $z \sim 15$, at which point the buildup of metal-enriched Pop II stars takes over (Section 3.1).
2. Stellar birth in the high-redshift Universe shares the same complexity as present-day star formation. Primordial gas is highly susceptible to fragmentation, and we expect the first stars to form in binary or higher-order multiple stellar systems with a wide range of separations, a flat mass ratio distribution, and a roughly thermal spread of orbital eccentricities (Section 3.2.2).
3. Most primordial stars build up in isolated high-redshift halos, which have not been affected by stellar feedback. We call this the standard pathway of Pop III star formation. We expect a wide distribution of masses, potentially ranging from the subsolar regime up to several hundred solar masses. The initial mass function (IMF) is approximately logarithmically flat, which makes it top heavy compared to the Milky Way values (Section 3.2.3).
4. Feedback from the first massive Pop III stars plays a major role in regulating subsequent star formation. Photoionization and supernovae (SNe) dominate within individual halos, whereas the Lyman–Werner and X-ray backgrounds built up by Pop III stars and their remnants govern the evolution on cosmic scales (Sections 4.2 and 4.3).
5. Pop III SNe produce metals that enrich gas associated with their host halo (internal enrichment) but can also reach neighboring halos (external enrichment). The time it takes Pop II stars to form from metal-enriched gas remains poorly constrained and depends on the halo mass, the SN energy, and details of the local environment. It is plausible that some fraction of the observed extremely metal-poor stars in the Milky Way and its satellites were enriched by single SNe (Section 4.4).
6. If they exist, Pop III stars with masses below $\sim 0.8\,M_\odot$ will have survived until today and should be detectable in surveys of extremely metal-poor stars. However, none has yet been found. At the high-mass end, observations point toward Pop III stars that end their lives in CCSNe, implying a mass range of 20–40 $M_\odot$. Evidence for more massive stars is sparse (Sections 5.1 and 5.2).
7. Under very rare and extreme conditions, the formation of SMSs is possible. Subjected to accretion rates of $\dot{M} \sim 0.1\,M_\odot\,\mathrm{year}^{-1}$ or higher, they can grow to masses of several $10^5\,M_\odot$, before the general relativistic instability kicks in and triggers the collapse to a black hole. These objects can be the seeds for the supermassive black holes observed in high-redshift quasars. Because the infalling gas is expected to fragment, runaway collisions in dense and deeply embedded stellar systems are thought to be important in this process (Section 3.4).

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FUTURE ISSUES

1. From a theoretical point of view, better understanding the impact of stellar feedback is a key challenge of current research into Pop III star formation. Specifically, the question of how photoionizing radiation escapes from the immediate vicinity of the star remains uncertain. On large scales, the relative importance of Lyman–Werner feedback, X-rays, and streaming velocities remains unclear, with several recent studies coming to highly different conclusions. Overall, it is likely that feedback has a negative effect on the Pop III star-formation rate, but its influence on the mass distribution and the multiplicity of Pop III stars remains undetermined.
2. On the observational side, no genuine low-mass Pop III star has been found so far in stellar archeological surveys. With the increasing sample size provided by future surveys, it will become possible to either confirm or rule out the existence of Pop III stars with masses below $\sim 0.8\,M_\odot$ with high statistical significance.
3. Halos forming Pop III stars are unlikely to be observable with the *James Webb Space Telescope* (JWST) without the aid of gravitational lensing, unless they have higher masses and star-formation efficiencies than current models predict. Supermassive Pop III stars or extremely strongly lensed systems should be bright enough to see, but will be very rare, and so it is plausible that none will be detected (Section 5.1.1). Pop III SNe should be visible with JWST, but are also very rare. Nevertheless, serendipitous discoveries remain a realistic possibility. Pop III gamma-ray bursts (GRB) are unlikely to be detected using current facilities, but will become accessible by next-generation GRB telescopes (Section 5.1.2). Much work still needs to be done to understand how to relate detections (or nondetections) of high-redshift Pop III stars to constraints on models of Pop III star formation.
4. Detection and characterization of the global 21-cm signal at high redshift will allow us to put stringent constraints on the properties of high-mass Pop III stars as well as the early evolution of the cosmic star-formation rate density.
5. The detection of gravitational waves from merging binary black holes in the mass range forbidden by pair-instability models provides additional support for the importance of stellar collisions, possibly occurring in the star clusters that are a natural consequence of the standard Pop III formation pathway (Section 5.3). Our ability to use these events to constrain the physics of Pop III star formation will improve as our sample size grows.

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